

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate

Monday 15 May 2023

Afternoon (Time: 1 hour 30 minutes)

Paper reference **31524H**

Sport
UNIT 1: Anatomy and Physiology

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION A: The Skeletal System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 1 shows the bones in the upper skeleton.

1 Identify the bones labelled A–D.

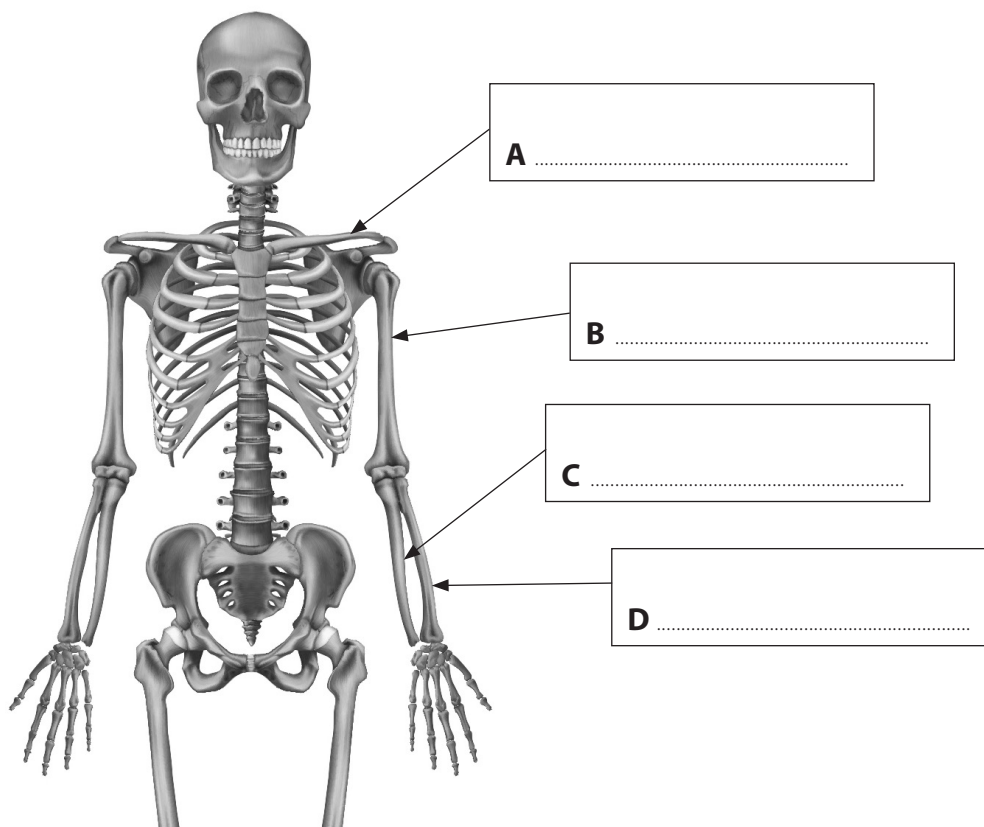


Figure 1

(Source: © studiovin/Shutterstock)

(Total for Question 1 = 4 marks)



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- (4)

[illegible]

- (1)

(Total for Question 2 = 5 marks)



Emily is a rugby player who specialises in kicking the ball. She relies heavily on her knee joint to perform the kicks successfully.



(Source: © Oleksandr Osipov/Shutterstock)

Figure 2

- 3** (a) State the movement that takes place in Emily's left knee joint (circled) when she performs the kick.

(1)



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(b) Explain how synovial fluid enables Emily to kick the rugby ball.

(3)

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(Total for Question 3 = 4 marks)

TOTAL FOR SECTION A = 13 MARKS



SECTION B: The Muscular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Table 1 shows the types of muscle and examples of where they are found.

4 Complete **Table 1** to identify the muscle type or where it is found as indicated **A–C**.

Muscle type	Where it is found
A	Heart
Smooth	B
C	Biceps

Table 1

(Total for Question 4 = 3 marks)

When a person is exercising one response of the muscular system is an increase in muscle pliability.

5 State **three other** responses of the muscular system when a person is exercising.

1

2

3

(Total for Question 5 = 3 marks)



One of the adaptations of the muscular system to exercise is an increased storage of fat.

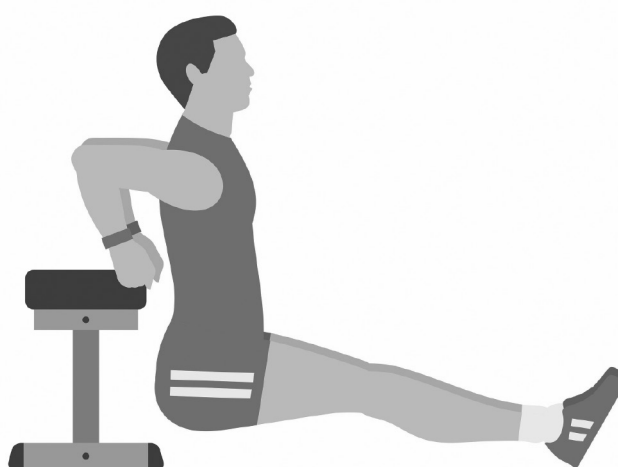
6 Explain why an increased storage of fat can benefit an endurance athlete.

(Total for Question 6 = 2 marks)



Mo is a basketball player.

As part of his training he participates in gym-based sessions, mainly doing body-weight exercises. One such exercise is the triceps dip. **Figure 3** shows him completing the downward phase of a triceps dip.



(Source: © VOLYK IEVGENII/Shutterstock)

Figure 3

- 7 (a) Describe the antagonistic muscle action that enables Mo to complete the downward phase of the triceps dip.

(2)



Over a period of time, Mo's muscular system has adapted to the training programme, causing muscular hypertrophy.

- (b) Explain the impact of hypertrophy of Mo's leg muscles on his basketball performance.

(3)

(Total for Question 7 = 5 marks)

TOTAL FOR SECTION B = 13 MARKS



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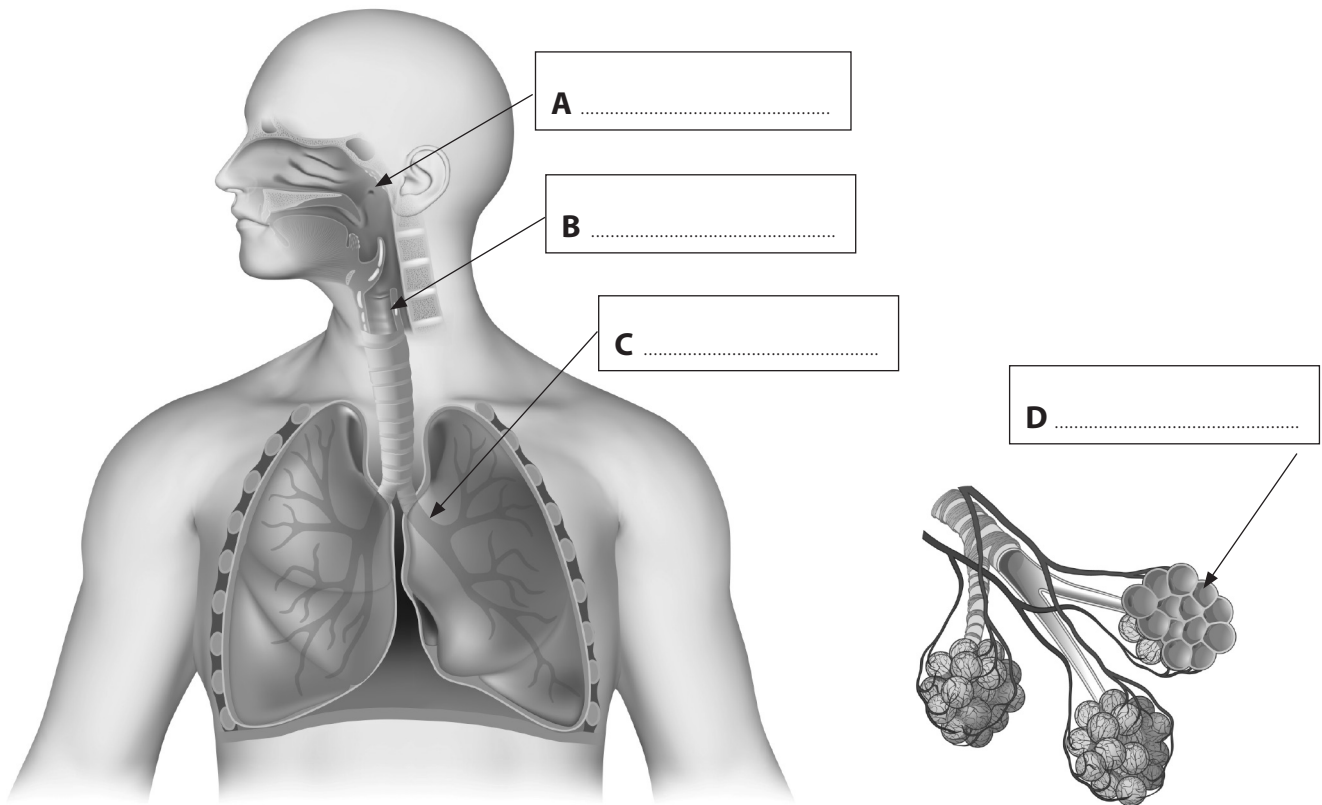


SECTION C: The Respiratory System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 4 shows the structure of the respiratory system.

8 Identify the respiratory structures labelled A–D.



(Source: © Alila Medical Media/Shutterstock)

(Source: © first vector trend/
Shutterstock)

Figure 4

(Total for Question 8 = 4 marks)



James is a road racing cyclist. During a race his tidal volume increases.

- 9 (a) Explain why James' tidal volume increases during his race.

(3)

James has been a cyclist for 15 years. During this time his respiratory system has adapted by increasing the strength of his respiratory muscles and increasing his oxygen and carbon dioxide diffusion rate.

- (b) Explain how an increased strength of the respiratory muscles affects James' cycling performance.

(3)



(c) Analyse the effect of an increase in the oxygen and carbon dioxide diffusion rate on James' cycling performance.

(6)

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(Total for Question 9 = 12 marks)

TOTAL FOR SECTION C = 16 MARKS



SECTION D: The Cardiovascular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Plasma is one component of blood. One function of plasma is to help regulate body temperature.

10 Complete **Table 2** by:

- (a) stating **two other** components of blood.
- (b) giving **one** function of each component of blood.

	Component of blood	Function of component of blood
Example	Plasma	Helps regulate body temperature
1	(1)	(1)
2	(1)	(1)

Table 2

(Total for Question 10 = 4 marks)

11 Describe hyperthermia.

.....

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.....

(Total for Question 11 = 2 marks)



Penny is a runner.

Figure 5 shows her heart rate before, during and after a race.

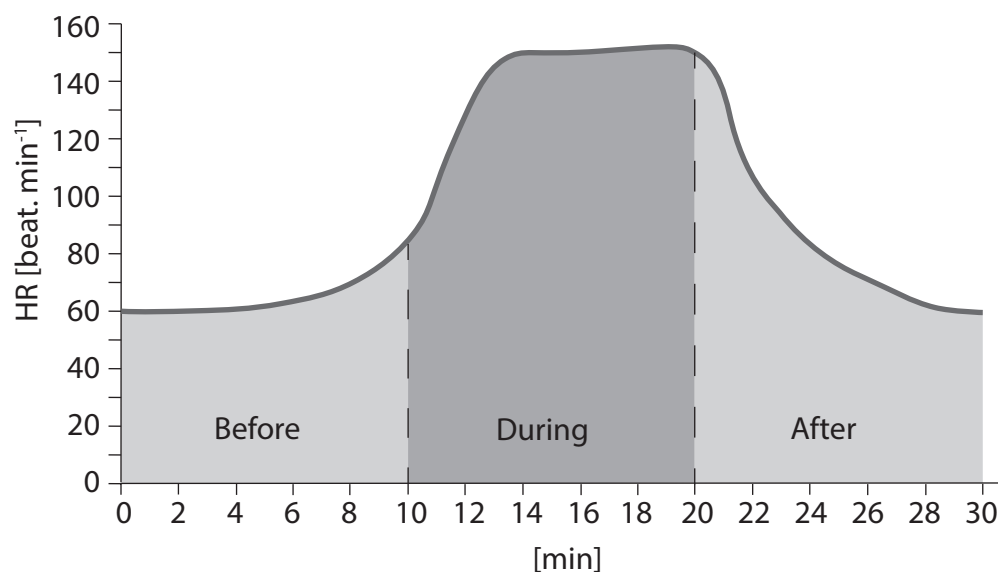


Figure 5

12 (a) Explain, using **Figure 5**, the changes to Penny's heart rate **before** the race.

(2)

(b) Explain, using **Figure 5**, the changes to Penny's heart rate **during** the race.

(3)

(Total for Question 12 = 5 marks)



Kellis is a netball player.

Figure 6 shows the percentage distribution of blood at rest and during a netball game.

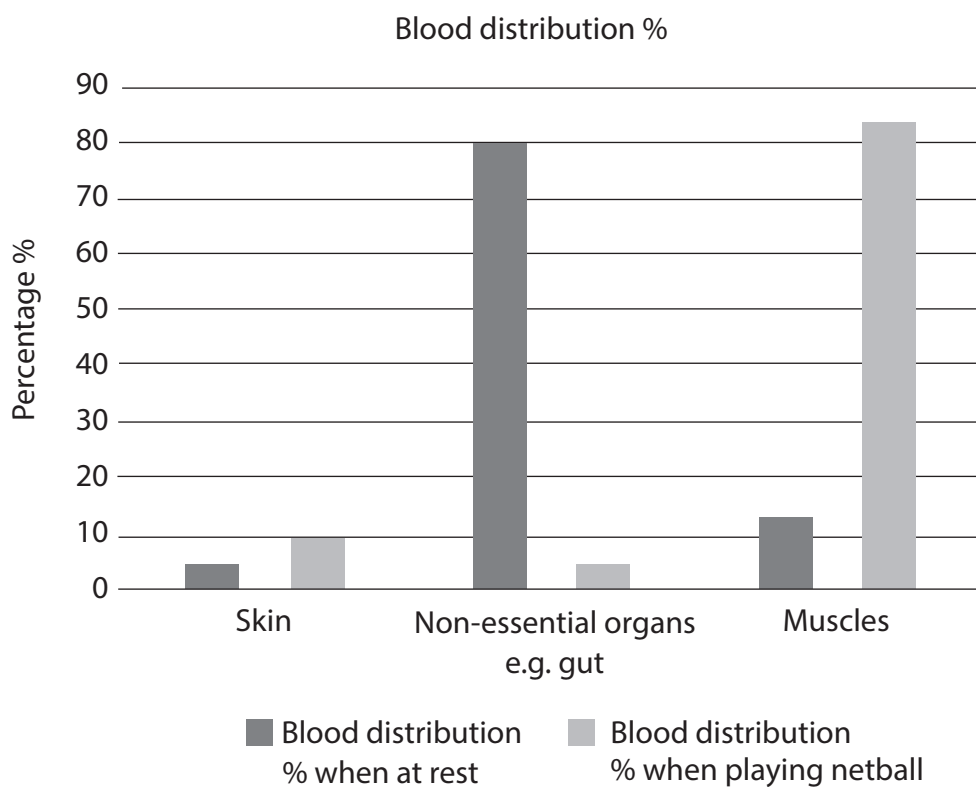


Figure 6

13 Analyse, using **Figure 6**, the changes in distribution of Kellis' blood flow to her skin, non-essential organs and muscles when playing netball.

(6)



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(Total for Question 13 = 6 marks)

TOTAL FOR SECTION D = 17 MARKS



SECTION E: Energy Systems for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

14 Complete Table 3 by:

- (a) stating the chemical source or fuel used for each energy system.
- (b) giving the amount of ATP produced by each energy system.

Energy system	Chemical source/fuel	Amount of ATP produced
ATP-PC	(1)	(1)
Lactate	(1)	(1)

Table 3

(Total for Question 14 = 4 marks)

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15 Describe the electron transport chain.

(Total for Question 15 = 3 marks)



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(Total for Question 16 = 6 marks)

TOTAL FOR SECTION E = 13 MARKS



P 7 4 8 6 7 A 0 2 1 2 4

SECTION F: Interrelationships between Body Systems for Sports Performance

Answer the question. Write your answer in the space provided.

Jesse is a footballer. He tested his fitness at the start and at the end of a three-month training programme. The tests showed significant improvements to Jesse's cardiovascular and energy system adaptations.

- 17** Analyse how the adaptations to Jesse's cardiovascular and energy systems will affect his fitness for football.

(8)

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(Total for Question 17 = 8 marks)

TOTAL FOR SECTION F = 8 MARKS
TOTAL FOR PAPER = 80 MARKS



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Mark Scheme (Results)

June 2023

Pearson BTEC Nationals
In Sport (31524H)
Unit 1: Anatomy and Physiology

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June 2023

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Unit 1: Anatomy and Physiology

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Mark
1	Award one mark for labelling each bone correctly. • A – Clavicle (1) • B – Humerus (1) • C – Ulna (1) • D – Radius (1) Accept phonetic spelling.	4

Question Number	Answer	Mark
2a	Award one mark for identifying the postural deviation and one mark for describing it. Two marks for each deviation. Kyphosis (1) curve and can cause a hunchback appearance. (1) Scoliosis (1) abnormal curvature to the left or right (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
2b	Award one mark for stating the classification of joint found at the thoracic vertebrae. • Cartilaginous/Slightly moveable (1)	1

Question Number	Answer	Mark
3a	Award one mark for movement occurring at the knee joint. • Extension	1

Question Number	Answer	Mark
3b	Award one mark for identifying synovial fluid lubricates the joint and up to two marks for linked expansion. Synovial fluid lubricates the joint (1) and reduces friction/prevents bones from rubbing together (1) to enable Emily to have a greater range of movement/power. (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
4	Award one mark for labelling each muscle type/example correctly. • A – Cardiac (1) • B – Intestines/stomach (1) • C – Skeletal (1)	3

Question Number	Answer	Mark
5	Award one mark for each response, up to three marks. • Increased blood supply (1) • Increased temperature (1) • Microtears (1) • Lactate build up (1)	3

Question Number	Answer	Mark
6	Award one mark for benefit and one mark for linked expansions. • Use fat as an (alternative) energy source (1) helps carbohydrate stores last longer in the race (1) Accept any other appropriate answer/example.	2

Question Number	Answer	Mark
7 (a)	Award two marks for identifying the antagonistic pair and describing their actions. The triceps is the agonist/contracts/working muscle (1) the biceps is the antagonist/relaxes to enable the agonist (triceps) to work. (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
7 (b)	Award one mark for explaining that muscular hypertrophy will be beneficial and up to two further marks for linked expansion. Increases the size/strength of his leg muscles (1) therefore he can create more force/power (1) enabling Mo to jump high to do a shot/defend (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
8	Award one mark for labelling each respiratory structure correctly. • A – Pharynx (1) • B – Larynx (1) • C – Bronchi (1) • D – Alveoli (1)	4

Question Number	Answer	Mark
9(a)	Award three marks for explaining why tidal volume increases. Due to an increased demand/need of O₂ within the muscles(1) therefore more O₂ taken in (1) CO₂ production has increased (1) therefore more CO₂ is removed (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
9 (b)	Award one mark for identifying the respiratory muscles and up to two additional marks for a linked expansion. Credit to a total of three marks. Diaphragm/intercostal muscles (increase in strength) (1) allowing expansion of the chest cavity/easier to take deeper breaths/more air/O₂ taken in (1) and more energy can be produced/delay fatigue/reduced lactate build up (1) Accept any other appropriate answer.	3

Question number	Indicative content	
9(c)	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • (General) Diffusion rate is the speed of transfer of gases • (General) Gases move from a high concentration to a low concentration • The steeper the diffusion gradient the quicker the gases move • An increased diffusion rate of oxygen would mean greater/more uptake of oxygen into the blood/delivery of oxygen to the muscles • The greater/more removal of carbon dioxide will prevent CO₂ build up Affects on Performance • (With more oxygen available) to the muscles they can work aerobically for longer/recover more quickly/break down lactic acid • so he can maintain his pace/(pedal) speed/high intensity and respond to people overtaking him if necessary/complete the race in a quicker time • The onset of fatigue will be delayed until later in the race Accept any other appropriate answer.	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

Question Number	Answer	Mark												
10a & 10b (Clip together)	Award one mark for each correct identification of the component of blood, up to a total of two marks. Award one mark for each associated function, up to a total of two marks. <table border="1"> <thead> <tr> <th></th><th>(a) - Component of blood</th><th>(b) - Function of component of blood</th></tr> </thead> <tbody> <tr> <td>1</td><td>Red blood cells/ RBC (1)</td><td>Transport/deliver/carry oxygen (1)</td></tr> <tr> <td>2</td><td>Platelets (1)</td><td>Clot the blood (1) / prevent bleeding (1)</td></tr> <tr> <td>3</td><td>White blood cells/WBC (1)</td><td>Fight infection (1)</td></tr> </tbody> </table>		(a) - Component of blood	(b) - Function of component of blood	1	Red blood cells/ RBC (1)	Transport/deliver/carry oxygen (1)	2	Platelets (1)	Clot the blood (1) / prevent bleeding (1)	3	White blood cells/WBC (1)	Fight infection (1)	2 + 2
	(a) - Component of blood	(b) - Function of component of blood												
1	Red blood cells/ RBC (1)	Transport/deliver/carry oxygen (1)												
2	Platelets (1)	Clot the blood (1) / prevent bleeding (1)												
3	White blood cells/WBC (1)	Fight infection (1)												

Question Number	Answer	Mark
11	Award one mark for identifying what hyperthermia is and up to one mark for linked expansion. - Elevated body temperature above 38°C/100F/abnormally high body temperature (1) - due to failed thermoregulation/inability to sweat (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
12 (a)	Award up to two marks for explaining the changes in heart rate before the race. There is an anticipatory rise (1) due to adrenaline being released (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
12 (b)	Award up to three marks for explaining the changes in heart rate during the race. There is a rapid increase in heart rate in the first three minutes (1) due to increased demand of O₂ (1) the heart rate then plateaus (1) because oxygen supply has met demand. (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
13	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • Redistribution of blood is called the vascular shunt happens through vasoconstriction which is reduced blood flow • Vasodilation which is increased blood flow,(increased blood flow) greater supply of oxygen and nutrients to working muscles • (increased blood flow) for thermoregulation to cool the body (via evaporation of sweat) • (less blood flow) as the priority is movement rather than digestion Impact on Performance • Enable Kellis to play at a high intensity for a longer duration • Help Kellis withstand fatigue until later in the game Accept any other answer	6
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

Question Number	Answer	Mark									
14	Award one mark for each correct identification of the chemical source, up to a total of two marks. Award one mark for the amount of ATP produced, up to a total of two marks. <table> <tr> <th>Energy system</th><th>Chemical source/fuel</th><th>Amount of ATP produced</th></tr> <tr> <td>ATP-PC</td><td>Phosphocreatine/PC/ creatine phosphate/CP (1)</td><td>1 ATP (1)</td></tr> <tr> <td>Lactate</td><td>Glycogen/glucose (1)</td><td>2 ATP (1)</td></tr> </table>	Energy system	Chemical source/fuel	Amount of ATP produced	ATP-PC	Phosphocreatine/PC/ creatine phosphate/CP (1)	1 ATP (1)	Lactate	Glycogen/glucose (1)	2 ATP (1)	4
Energy system	Chemical source/fuel	Amount of ATP produced									
ATP-PC	Phosphocreatine/PC/ creatine phosphate/CP (1)	1 ATP (1)									
Lactate	Glycogen/glucose (1)	2 ATP (1)									

Question Number	Answer	Mark
15	Award one mark for each descriptive point, up to a maximum of three marks. • (The electron transport chain/ETC) occurs when oxygen is present/part of the aerobic system (1) • It occurs in the mitochondria (1) • It receives hydrogen ions (1) • 32–34 ATP are created (1) • Water is the by-product (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
16	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. ATP-PC system • (General) System used for high intensity activity lasting up to 10 seconds/ Phosphocreatine stores are replenished in 2–4 minutes. • Used when hitting powerful shots (smash, drive, serve)/sprinting around the court to return shots • Breaks between points and games allow for recovery Lactate system • (General) Provides energy for moderate to high intensity activity, only lasts for 1–3 minutes • Will be used during long, intense rallies that last longer than 10 seconds • When recovery periods between points are short. Aerobic system • (General) Submaximal system (low/medium intensity) in exercise longer than 3 minutes/2-3 hours. • Used throughout the game to support the anaerobic systems • Fatigue later in the match/last the duration of the match • Works when intensity falls between points/games and helps the anaerobic systems/muscles recover. Accept any other appropriate answer.	6
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Provides little or no reference to the context in the question. • A conclusion may be presented but will be generic and the supporting evidence will be limited. Limited attempt to address the question. • Response is likely to lack clarity, organisation and the required technical language.
Level 2	3–4	• Demonstrates accurate knowledge and understanding. • Line(s) of argument occasionally supported through the application of relevant references to context in question.

		• Judgement is made from a partially developed discussion, although the discussion may be imbalanced or superficial in places. Learners will produce some statements with development in the form of mostly accurate and relevant factual material leading to an assessment being presented. • The response may contain parts which lack clarity or organisation. There is evidence of correct technical language being used.
Level 3	5-6	• Demonstrates accurate knowledge and understanding. • Line(s) of argument supported throughout by sustained application of relevant references to context in the question. Might demonstrate the ability to integrate and synthesise relevant systems. • Arrives at a supported judgement from a well-developed and logical balanced discussion, containing logical chains of reasoning. Demonstrates an awareness of competing arguments using these to reach a valid assessment. • Response demonstrates good organisation, clarity and use of technical language.

Question Number	Answer	Mark
17	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Cardiovascular system - Jesse's cardiovascular system will have adapted through cardiac hypertrophy - this increases stroke volume - giving him a greater maximal cardiac output - this will give Jesse a greater heart rate range to utilise when playing football - Jesse will also have an increased blood volume Capillarisation around the alveoli and in the muscles will increase the efficiency of gaseous exchange. - More oxygen is transported to his muscles Decreased heart rate recovery time Energy systems Increased creatine stores, providing more energy and supporting recovery from high intensity runs around the pitch Increased tolerance to lactate will enable Jesse to clear out the lactate more efficiently due to larger oxygen consumption Increased use of fats as an energy source, so that glycogen can be used later in the game Increased storage of glycogen to provide energy for ATP production Increased number of mitochondria will increase the amount of ATP production to support an increased demand for energy - More efficient energy systems. Impact on performance - He can work at a higher intensity for longer, e.g. can make high intensity runs around the pitch - Keep active around the pitch for longer before he starts feeling fatigued/will last the 90 minutes. Accept any other appropriate answer.	8

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	- Demonstrates isolated elements of knowledge and understanding. - Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question.

		Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	4-6	- Demonstrates some accurate knowledge and understanding. - Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. - Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	7-8	- Demonstrates mostly accurate knowledge and understanding. - Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. - Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Learner Registration Number				

Pearson BTEC Level 3 Nationals Certificate

Time 1 hour 30 minutes

Paper reference **31524H**

Sport
UNIT 1: Anatomy and Physiology

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
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Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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Turn over ►

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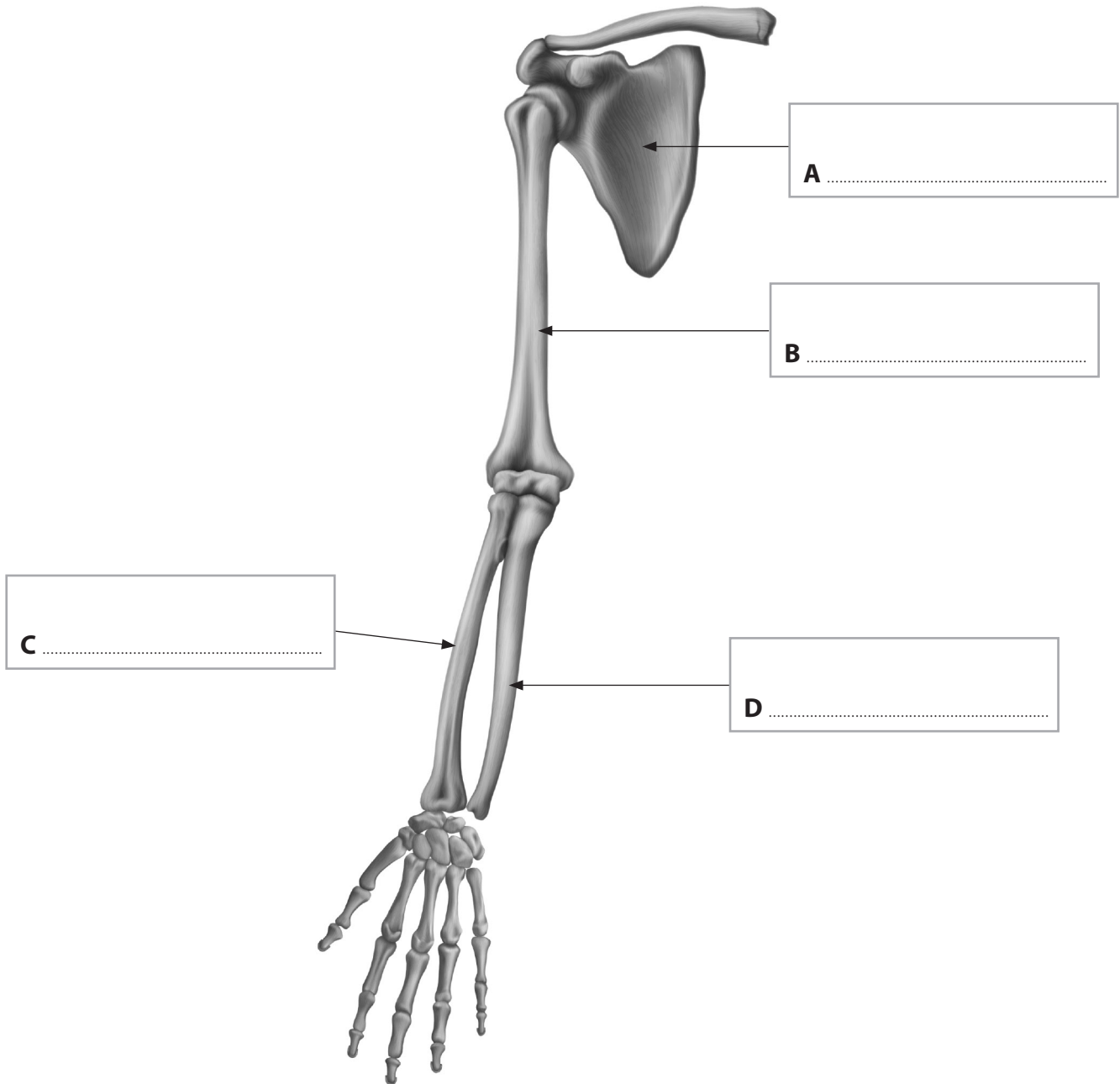
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Section A: The Skeletal System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

- 1 (a) Identify the bones labelled A–D.

(4)



(Source: © Shutterstock)

Figure 1

- (b) State the type of synovial joint that is found at the shoulder.

(1)

(Total for Question 1 = 5 marks)



Zara is a badminton player. During a game she regularly runs, jumps and changes direction.

- 3 Explain why regular participation in weight-bearing activities helps to reduce the likelihood of osteoporosis.

(Total for Question 3 = 3 marks)

TOTAL FOR SECTION A = 13 MARKS

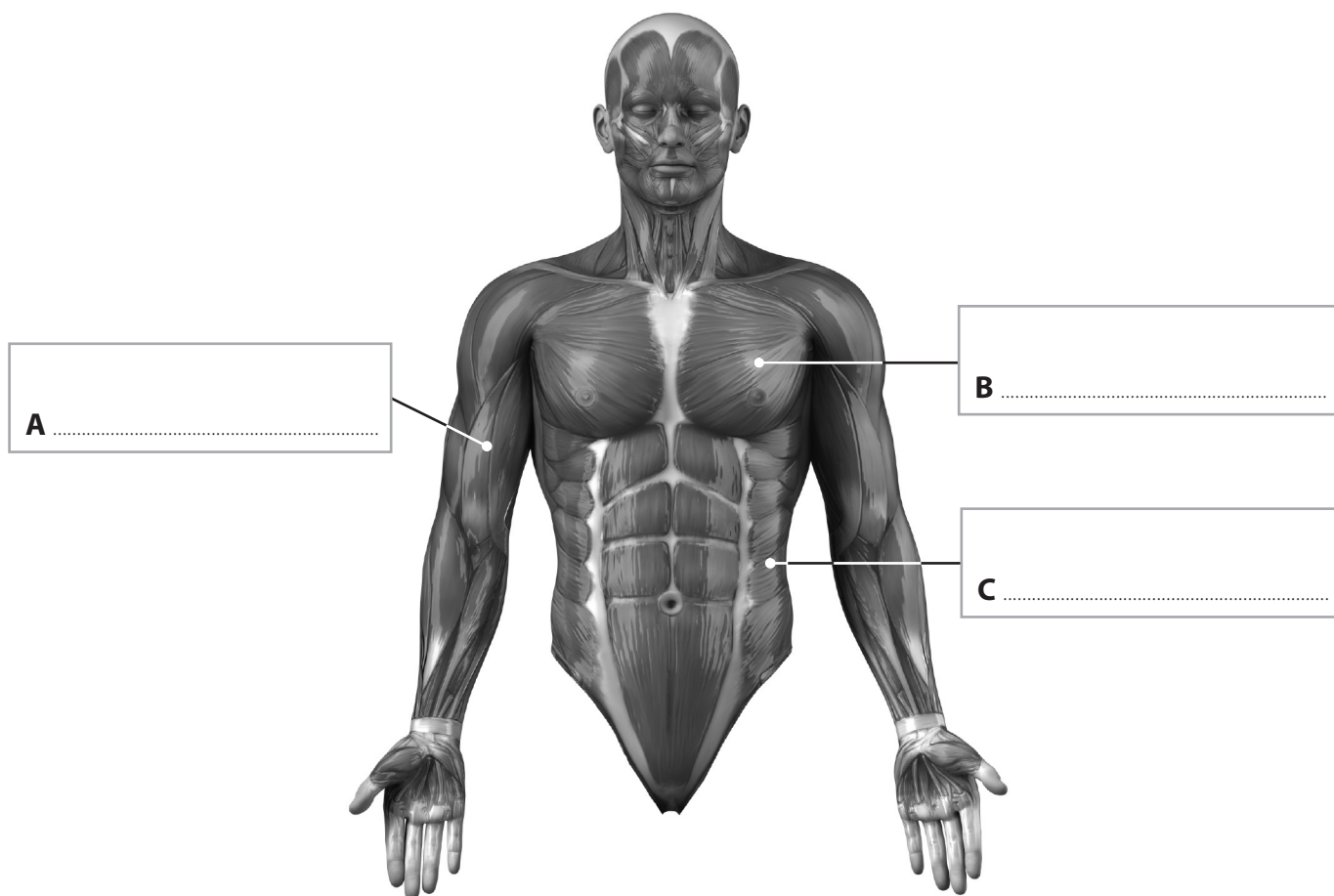


SECTION B: The Muscular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 2 shows the muscles in the upper body.

4 Identify the muscles labelled **A–C**.



(Source: © Shutterstock)

Figure 2

(Total for Question 4 = 3 marks)

5 State **three** types of skeletal muscle fibre.

- 1
- 2
- 3

(Total for Question 5 = 3 marks)



6 Describe, using an example of a sporting action, the role of an agonist muscle.

(Total for Question 6 = 2 marks)

Sundeep is a 400 m runner.

As part of his training regime he participates in two track sessions per week. One response that occurs during a training session is that his muscles increase in temperature.

7 (a) Explain how increased muscle temperature will affect Sundeep when running.

(2)

Over a period of time, Sundeep's muscular system has adapted to the training programme, causing an increased tolerance to lactate.

(b) Explain the impact an increase in tolerance to lactate has on Sundeep's 400 m running performance.

(3)

(Total for Question 7 = 5 marks)

TOTAL FOR SECTION B = 13 MARKS

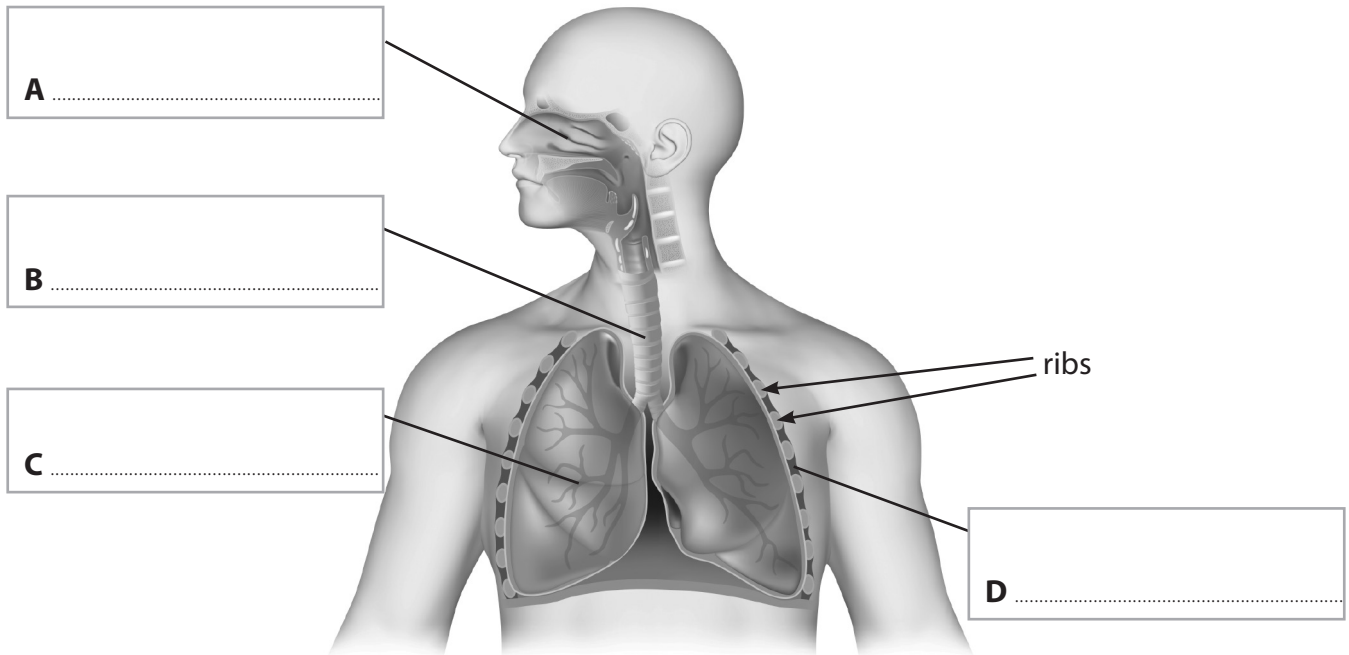


SECTION C: The Respiratory System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 3 shows the structure of the respiratory system.

8 Identify the respiratory structures labelled A–D.



(Source: © Shutterstock)

Figure 3

(Total for Question 8 = 4 marks)



Ethan is a 1500 m swimmer.

- 9 (a) Explain **two** immediate respiratory responses for Ethan when he starts swimming in a 1500 m race.

(4)

- (b) Describe how the chemical control of breathing changes Ethan's minute ventilation (VE) during the swim.

(2)



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TOTAL FOR SECTION C = 16 MARKS



Turn over ►

SECTION D: The Cardiovascular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 4 is an incomplete flow diagram of the structures in the heart that control the cardiac cycle.

10 Identify the **two** structures needed to complete the flow diagram in **Figure 4**.

A



Atrioventricular node (AVN)



B



Purkinje fibres

Figure 4

(Total for Question 10 = 2 marks)

11 Describe the role of the Purkinje fibres.

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(Total for Question 11 = 2 marks)



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12 Describe hypothermia.

(Total for Question 12 = 2 marks)



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- (1)

- (c) Analyse how the sympathetic nervous system controls the changes in Ashley's cardiac cycle during her tennis match, and the impact of this control on her performance.

(6)

(Total for Question 13 = 11 marks)

TOTAL FOR SECTION D = 17 MARKS



SECTION E: Energy Systems for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Luke is a decathlete. He participates in a number of different events across two days. These events use all of his energy systems.

On day one, the high jump is one of these events.



(Source: © Shutterstock)

Figure 5

14 Explain why the ATP-PC energy system is used when participating in the high jump.

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(Total for Question 14 = 3 marks)



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On day two, the last event is the 1500 m run. Luke completes the run in 4 minutes and 25 seconds.

- 16** Evaluate the contributions of each of the **three** energy systems throughout Luke's 1500 m run.

(Total for Question 16 = 6 marks)

TOTAL FOR SECTION E = 13 MARKS



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SECTION F: Interrelationships between Body Systems for Sports Performance

Answer the question. Write your answer in the space provided.

Figure 6 shows Sammy in action during a game of football.



Figure 6

- 17** Analyse how the muscular system and the skeletal system work together at the hip, knee and ankle joints to allow Sammy's right leg, shown shaded in **Figure 6**, to move from Position A to Position B to strike the football.

(8)

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(Total for Question 17 = 8 marks)

TOTAL FOR SECTION F = 8 MARKS
TOTAL FOR PAPER = 80 MARKS



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Mark Scheme (Results)

January 2023

Pearson BTEC Nationals
In Sport (31824H)
Unit 1: Anatomy and Physiology

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January 2023

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Unit 1: Anatomy and Physiology

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.

Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Mark
1a General	Award one mark for labelling each bone correctly. • A – Scapula (1) • B – Humerus (1) • C – Radius (1) • D – Ulna (1) Accept phonetic spelling.	4

Question Number	Answer	Mark
1b General	Award one mark for type of joint found at the shoulder. • Ball and socket Accept phonetic spelling.	1

Question Number	Answer	Mark
2a Expert	Award one mark for identifying why flat bones and one mark for identifying why long bones will help when playing in the match and one mark for justifying each reason. Flat bones protect the vital organs (1) reducing the risk of injuries when being tackled. (1) Long bones enable leverage/movement to take place (1) to enable Warren to run with the ball. (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
2b General	Award one mark for stating a type of bone found in the skeleton. • Short (1) • Sesamoid (1) • Irregular (1) Accept any other appropriate answer.	1

Question Number	Answer	Mark
3 Expert	Award one mark for identifying that bones become stronger/denser and up to two marks for justifying that reason. The weight-bearing activities cause additional stress or damage to the bone/help make the bones stronger/denser. (1) The process happens through ossification (1) osteoclasts clear away old bone (1) increasing osteoblast activity to build new bone (1) allowing the bones to take up more calcium/minerals (1) and making them less likely to break/fracture in later life. (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
4 General	Award one mark for labelling each muscle correctly. • A – Bicep (1) • B – Pectorals (1) • C – Obliques (1) Accept phonetic spelling. *DNA = Do not accept	3

Question Number	Answer	Mark
5 General	Award one mark for each type of muscle contraction, up to three marks. • Type I(1) • Type IIa (1) • Type IIx (1)	3

Question Number	Answer	Mark
6 Graduate	Award up to one mark for describing the role of an agonist muscle and one mark for the sporting example of the action. • Muscle that contracts for the movement to take place (1) quadricep when extending the knee, kicking a ball (1) Accept any other appropriate answer/example.	2

Question Number	Answer	Mark
7 (a) Graduate	Award one mark for identifying how increased muscle temperature will affect him and one mark for linked expansion. Increases the speed/force of contraction/ increases the pliability of his muscles/increases the amount that his muscles can stretch/makes him more flexible (1) increases his stride length/reduces the risk of injuries (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
7 (b) Expert	Award one mark for explaining an increase in tolerance to lactate and up to two further marks for linked expansion. More efficient in clearing waste products (1) enabling him to work at a higher intensity for longer duration/run a faster time (1) fatigue is delayed/reduced until later in the race. (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
8 General	Award one mark for labelling each respiratory structure correctly. • A – Nasal cavity (1) • B – Trachea(1) • C – Bronchiole(s) (1) • D – Intercostal (muscles) (1) Accept phonetic spelling.	4

Question Number	Answer	Mark
9(a) Expert	Award up to four marks for explaining the effects on the respiratory system of taking part in the race. Two marks are awarded for increased breathing rate and two marks awarded for increased tidal volume. • Increased breathing rate/breathe quicker (1) due to a greater demand for oxygen in the race. (1) • Increased tidal volume/depth of breathing/more air (1) to meet the need of the body to remove waste products. (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
9 (b) Graduate	Award up to two marks for describing how chemical control changes minute ventilation. • Chemoreceptors detect an increase in CO₂/decrease in O₂/decrease in pH(1) • which increases minute ventilation. (1) Accept any other appropriate answer.	2

Question number	Indicative content	
9(c) Expert	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. - Gas exchange occurs through diffusion, where gases move from a high concentration to a low concentration Muscle - During the race there is a high PPO₂ in the blood/capillary - There is a low PPO₂ in the muscle - O₂ diffuses from the blood into the muscle - During the race there is a high PPCO₂ in the muscle - There is a low PPCO₂ in the blood - CO₂ diffuses from the muscle into the blood. Lungs - During the race there is a high PPO₂ in the alveoli - There is a low PPO₂ in the blood/capillary - O₂ diffuses from the alveoli into the blood/capillary - During the race there is a high PPCO₂ in the blood/capillary - There is a low PPCO₂ in the alveoli - CO₂ diffuses from the blood/capillary into the alveoli. Impact - Steeper diffusion gradient created - Enables more oxygen to be used in the tissue and the body - Gas exchange becomes more efficient and more CO₂ removed. Accept any other appropriate answer.	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	- Demonstrates isolated elements of knowledge and understanding. - Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. - Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	- Demonstrates some accurate knowledge and understanding. - Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. - Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.

Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.
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Question Number	Answer	Mark
10 General	Award one mark for identifying each structure correctly. • A – Sinoatrial node/SA node/SAN (1) • B – Bundle of His (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
11 Graduate	Award up to two marks for describing the role of Purkinje fibres. • Carry the impulse around the ventricles (1) causing them to contract (1)	2

Question Number	Answer	Mark
12 Graduate	Award one mark for identifying what hypothermia is and up to one mark for linked expansion. • Body temperature is below 35°C (1) which causes shivering/confusion/heart stopping (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
13 (a) Expert	Award one mark for explaining why Ashley's heart rate increases before she starts to exercise and three marks for linked expansion. This is an anticipatory rise (1) due to the release of adrenaline (1) this in turn will increase cardiac output/increase blood supply to working muscle/body (1) and enable more oxygen to be delivered to the working muscles (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
13 (b) General	Award one mark for stating the correct blood vessel that returns blood. • Vena cava (1) Accept any other appropriate answer.	1

Question Number	Answer	Mark
13 (c) Expert	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Sympathetic • Sympathetic nervous system increases heart rate • Messages sent down the sympathetic/accelerator nerve from the medulla oblongata • Which increases the firing rate/number of impulses of the sinoatrial node/SA node/SAN • causing the atria to contract more/with more force • the impulses are transferred causing more often/forceful contractions of the ventricles • To enable more blood/oxygen to be delivered to the working muscles during exercise Impact • to enable her to maintain/work at a higher intensity (e.g.) repeated sprints within a rally, whilst withstand/delaying fatigue	6
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

Question Number	Answer	Mark
14 Expert	Award one mark for identification of why the system is used and one mark for linked expansion up to three marks. It provides/releases energy quickly/immediate source of energy (1) so Luke can produce an explosive movement/so it can provide enough energy to complete the movement (1) the high jump/energy system lasts less than 10 seconds (1) ATP is resynthesised quickly for the next jump/recovery time is short. (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
15 Expert	Award up to four marks for describing anaerobic glycolysis. - Anaerobic glycolysis will provide the energy required (1) - Glycogen/glucose is broken down (1) into pyruvic acid (1) - Produces 2 ATP (1) - Lactic acid produced (as there is no oxygen) (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
16 Expert	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • The ATP-PC may be used at the start of the race or when sprinting. • The ATP-PC system is unable to supply enough energy for long enough due to it being a long duration activity • Lactic Acid system used to provide further energy for when increasing/maintain higher pace during the race • Lactic acid system cannot provide energy for long enough due to the build up of lactic acid • In the 1500 m Luke will predominately use the aerobic energy system to create ATP continuously • Because the race is long duration/to sustain energy production	6
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Provides little or no reference to the context in the question. • A conclusion may be presented, but will be generic and the supporting evidence will be limited. Limited attempt to address the question. • Response is likely to lack clarity, organisation and the required technical language.
Level 2	3–4	• Demonstrates accurate knowledge and understanding. • Line(s) of argument occasionally supported through the application of relevant references to context in question. • Judgement is made from a partially-developed discussion, although the discussion may be imbalanced or superficial in places. Learners will produce some statements with development in the form of mostly accurate and relevant factual material leading to an assessment being presented. • The response may contain parts which lack clarity or organisation. There is evidence of correct technical language being used.
Level 3	5–6	• Demonstrates accurate knowledge and understanding. • Line(s) of argument supported throughout by sustained application of relevant references to context in the question. Might demonstrate the ability to integrate and synthesise relevant systems.

		• Arrives at a supported judgement from a well-developed and logical balanced discussion, containing logical chains of reasoning. Demonstrates an awareness of competing arguments using these to reach a valid assessment. • Response demonstrates good organisation, clarity and use of technical language.
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Question Number	Answer	Mark
17 Expert	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • Hip is a ball and socket joint, made up of the femur and pelvis, • the movement taking place is flexion • the agonist muscle is the hip flexor and the antagonist is the gluteus maximus • the hip flexor (iliopsoas) is contracting concentrically and the gluteals are relaxing • Knee is a hinge joint, made up of the femur, tibia, • the movement taking place is knee extension • the agonist muscle is the quadriceps and the antagonist is the hamstrings • the quadriceps are contracting concentrically and the hamstrings are relaxing • Ankle joint is a hinge joint, made up of tibia, fibula, tarsals • movement is plantar flexion • the agonist muscle is the gastrocnemius and the antagonist is the tibialis anterior • the gastrocnemius is contracting concentrically and the tibialis anterior is relaxing • Skeletal muscles attach to bones via tendons and when they contract they pull on the bone producing movement when Sammy shoots • Ligaments provide stability for the joints, so that they do not dislocate • The muscle fibres being used within the shot are type IIX, which can generate a maximum force but only last a short duration	8

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question.

		Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	4-6	- Demonstrates some accurate knowledge and understanding. - Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. - Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	7-8	- Demonstrates mostly accurate knowledge and understanding. - Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. - Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.



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Candidate surname

Other names

Centre Number

Learner Registration Number

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Pearson BTEC Level 3 Nationals Certificate

Time 1 hour 30 minutes

Paper
reference

31524H

Sport

UNIT 1: Anatomy and Physiology

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

SECTION A: The Skeletal System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

- 1** (a) Name the **three** bones that form the elbow joint.

(3)

1

2

3

- (b) State the **type** of synovial joint that allows the head to rotate.

(1)

.....

(Total for Question 1 = 4 marks)

The skeletal system has a number of types of bone. Two of these types are long and short bones.

- 2** (a) State **two other** types of bone found in the skeletal system.

(2)

1

2

Shantel is a goalkeeper in a football team. Her long and short bone types help her to perform successfully in a match.

- (b) Explain how **long** and **short** bones help Shantel when goalkeeping in a match.

(4)

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(Total for Question 2 = 6 marks)



One of the additional factors that can affect the skeletal system is arthritis.

3 Explain how arthritis affects the skeletal system.

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(Total for Question 3 = 3 marks)

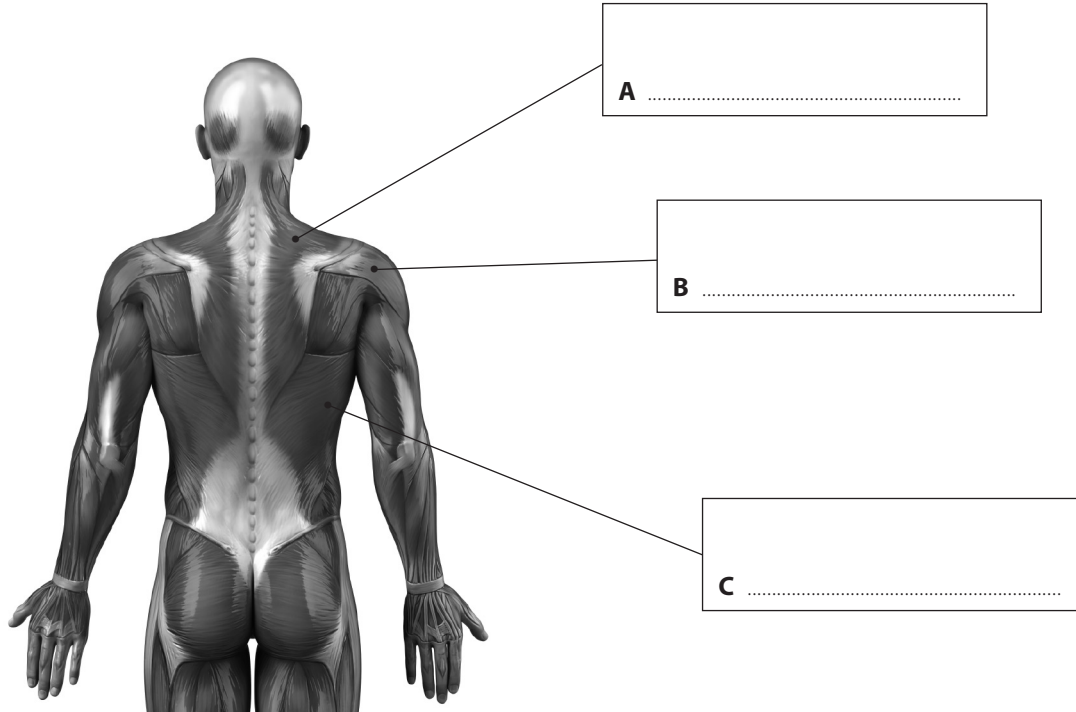
TOTAL FOR SECTION A = 13 MARKS



SECTION B: The Muscular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

4 Identify the muscles labelled A–C in Figure 1.



Source: © CLIPAREA | CUSTOM MEDIA/Shutterstock

Figure 1

(Total for Question 4 = 3 marks)

5 State **three** types of muscle contraction.

- 1
- 2
- 3

(Total for Question 5 = 3 marks)

6 Describe the **all or none law** of muscle contraction.

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-

(Total for Question 6 = 2 marks)



Tristan is a long-distance runner. One response that occurs during a training run is that his muscles become more pliable.

- 7 (a) Explain why increased muscle pliability will benefit Tristan when running.

(2)

Over a long period of time Tristan's muscular system has adapted to the training programme, causing an increase in myoglobin stores.

- (b) Explain the impact of an increase in myoglobin stores on Tristan's performance in a 10,000 m race.

(3)

(Total for Question 7 = 5 marks)

TOTAL FOR SECTION B = 13 MARKS



P 6 7 5 1 1 A 0 5 1 6

SECTION C: The Respiratory System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Alex is an endurance athlete who is starting altitude training.

- 8 Explain the negative effects on Alex's respiratory system when starting training at altitude.

(Total for Question 8 = 4 marks)

Asthma is a condition that affects the respiratory system.

- 9 Describe the condition asthma.

(Total for Question 9 = 2 marks)

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Nancy is a netball player. During a match Nancy needs to remove waste products from her body. One way this is supported is through chemical control of breathing.

- 10** (a) Explain how chemical control of breathing helps Nancy to remove waste products from her working muscles.

(4)

When Nancy is playing a netball match, her lung volumes change.

- (b) Analyse the changes to Nancy's lung volumes and the impact this will have on her netball performance.

(6)



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(Total for Question 10 = 10 marks)

TOTAL FOR SECTION C = 16 MARKS

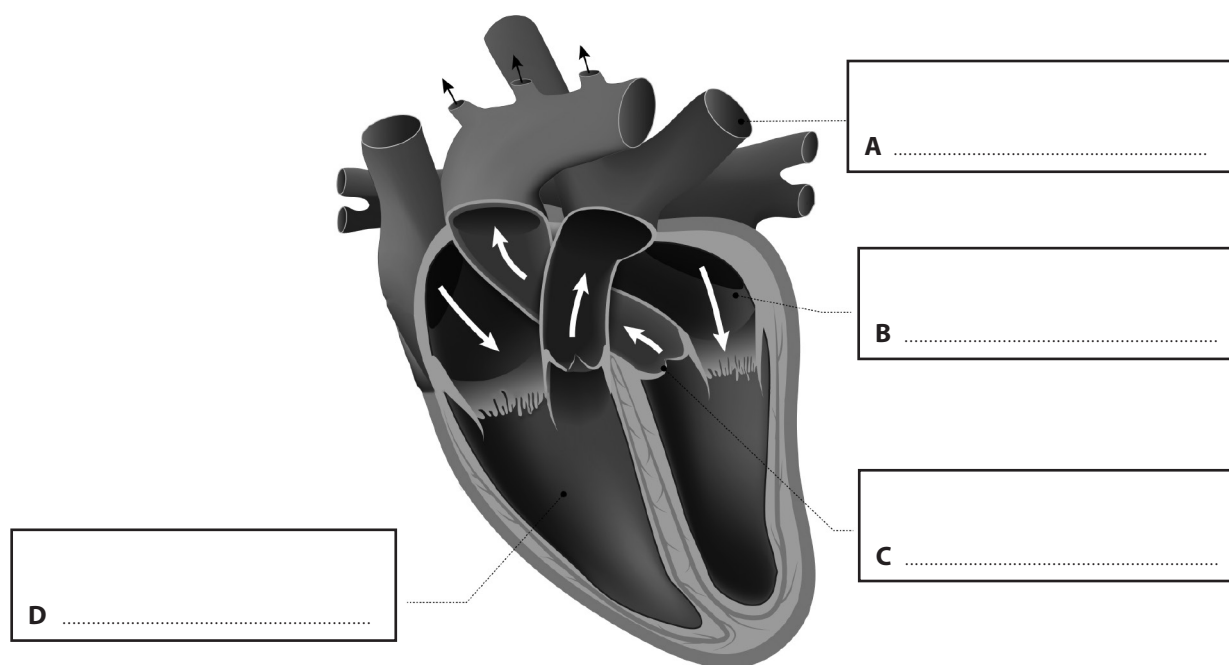


SECTION D: The Cardiovascular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 2 shows the heart.

11 Identify the structures of the heart labelled A–D.



© Source: Designua/Shutterstock

Figure 2

(Total for Question 11 = 4 marks)

12 Describe the **function** of capillaries.

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(Total for Question 12 = 2 marks)



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(Total for Question 13 = 11 marks)

TOTAL FOR SECTION D = 17 MARKS



SECTION E: Energy Systems for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

The aerobic energy system can be broken down into three processes.

14 State each of the **three** aerobic energy system processes in the order they occur.

1

2

3

(Total for Question 14 = 3 marks)

Rhea is a hockey player. Due to her training regime her aerobic system has adapted. One of these adaptations is that Rhea can now use fats as an energy source quicker.

15 Explain **two other** adaptations that have taken place within Rhea's aerobic energy system and the impact they have on her performance.

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(Total for Question 15 = 4 marks)

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16 Analyse the role of the **anaerobic** energy systems in making sure that Rhea is successful in the hockey match.

TOTAL FOR SECTION E = 13 MARKS



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(Total for Question 17 = 8 marks)

TOTAL FOR SECTION F = 8 MARKS
TOTAL FOR PAPER = 80 MARKS



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Mark Scheme (Results)

June 2022

Pearson BTEC Nationals
In Sport (31524H)
Unit 1: Anatomy & Physiology

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June 2022

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Unit 1: Anatomy & Physiology

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
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Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

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- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Mark
1a	Award one mark for naming each bone correctly. Humerus (1) Ulna (1) Radius (1) Accept phonetic spelling.	3

Question Number	Answer	Mark
1 (b)	Award one mark for stating a joint that allows the head to rotate. Pivot Accept phonetic spelling.	1

Question Number	Answer	Mark
2a	Award one mark for stating a type of bone found in the skeleton for a total of two marks. - Flat (1) - Sesamoid (1) - Irregular (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
2b	Award one mark for identifying why long bones and one mark for identifying why short bones will help when playing in the match and one mark for each linked expansion. Long bones enable movement to take place (1) to enable Shantel to move across the goal to make a save (1) Short bones absorb the shock/weight bearing (1) reducing the risk of injuries when saving a shot/allows her to push up off the ground to get back up after a save (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
3	Award up to three marks for explaining arthritis. Arthritis is when there is an inflammation (in the joint) (1) due to the breakdown/wearing away of cartilage (1) causing pain/stiffness/friction (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
4	Award one mark for labelling each muscle correctly. • A – Trapezius (1) (DNA traps) • B – Deltoid (posterior) (1) • C – Latissimus dorsi (1) (DNA lats) Accept phonetic spelling. *DNA = Do not accept	3

Question Number	Answer	Mark
5	Award one mark for each type of muscle contraction, up to three marks. • Concentric/isotonic (1) • Eccentric/isotonic (1) • Isometric (1)	3

Question Number	Answer	Mark
6	Award up to two marks for describing the all or none law. For a motor unit to contract (1) it must receive an impulse/signal (1) all muscle fibres must contract/none of the muscles fibres contract (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
7 (a)	Award one mark for identifying how increased muscle pliability will benefit him and one mark for a linked expansion. Increase the amount that his muscles can stretch/make him more flexible (1) increase his stride/reduce the risk of injuries (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
7 (b)	Award one mark for explaining an increase in myoglobin stores and up to two further marks for a linked expansion. Enables him to store more oxygen in his muscles which will enable him to transport more oxygen to the mitochondria (1) have more energy/make more energy aerobically (1) and run a faster time/fatigue will set in later in the race (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
8	Award one mark for explaining the effects on an endurance athlete from training at altitude for a total of four marks. Air is at a lower pressure/less oxygen in air (1) which causes hypoxia/body (1) the respiratory system works harder by increasing breathing rate/shortness of breath (1) this can result in lower capacity to train/reduced performance fatigue quicker/altitude sickness (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
9	Award two marks for describing asthma. Where the airways become restricted (1) making it harder to breathe/wheezing/coughing/shortness of breath (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
10a	Award up to four marks for explaining chemical control of breathing enables removal of waste products. Chemoreceptors (1) detect increased levels of carbon dioxide/lower pH (1) (send messages) to the medulla oblongata (1) therefore this speeds up/increases breathing rate (to remove carbon dioxide) (1)	4

Question number	Indicative content	
10(b)	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Lung Volumes changes and effects • Her tidal volume will also increase, meaning that she will be breathing deeper • Minute ventilation/pulmonary ventilation will increase • Respiratory rate x tidal volume = minute ventilation • Her inspiratory reserve volume/IRV will decrease meaning that more oxygen can reach the alveoli • Due to this Nancy’s expiratory reserve volume/ERV will decrease meaning that she can breathe out more carbon dioxide • residual volume/vital capacity/total lung volume do not change • More oxygen/air taken in, so more oxygen will go to the working muscles/remove more waste products • To prevent the build up lactic acid/sustain aerobic energy production Impact on performance • So that she can work at a higher intensity for longer e.g. can maintain movement around the court/dodging. • The delay of fatigue will come later in the game, (e.g.) so that Nancy can maintain intensity of performance in the 4th quarter. Accept any other appropriate answer.	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding.

		• Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.
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Question Number	Answer	Mark
11	Award one mark for naming each structure correctly. • A – Pulmonary artery (1) • B – Left atrium (1) • C – Aortic/semi-lunar valve (1) • D – Right ventricle (1)	4

Question Number	Answer	Mark
12	Award up to two marks for describing the function of capillaries. Connect arteries and veins/arterioles and venules(1) they provide site for gas exchange (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
13 (a)	Award two marks for explaining the impact of increased blood volume and two marks for a linked justification. Stroke volume/SV is the amount of blood pumped from the heart per beat (1) therefore more oxygen is delivered to the working muscle (1) therefore enabling Robin to play at a higher intensity for a longer duration (1) and delay the onset of fatigue/recovery quicker between points (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
13 (b)	Award one mark for stating the component of blood that would carry most oxygen. • Red blood cells/rbc (1)	1

Question Number	Answer	Mark
13 (c)	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Functions of the cardiovascular system • Delivering oxygen and nutrients, to be transported to Robin's working muscles • Enables an increase in aerobic energy production • Removing waste products – carbon dioxide and lactate from the tissue to avoid build up at the muscle • Thermoregulation occurs due to vasodilation of blood vessels near to the skin • Which carries blood nearer to the skin's surface, so that excess heat can be lost. This ensures that Robin does not overheat during the game Impact on Performance • Meaning he can work at higher intensity for longer • Fatigue will occur later • (e.g) he can still run around the court within the game	6
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
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Question Number	Answer	Mark
14	Award one mark for each process in the correct order. • 1 – Aerobic glycolysis (1) • 2 – Krebs cycle/citric acid cycle(1) • 3 – Electron transport chain/ETC (1) Accept phonetic spelling.	3

Question Number	Answer	Mark
15	Award one mark for each aerobic adaptation and one mark for a linked expansion/justification. • Increased storage of glycogen (1) enabling Rhea to keep running at a high intensity in the latter stages of the game/be able to still attack and defend late in the game/delays fatigue (1) • Increased number of mitochondria (1) increases the ability to produce more energy/more aerobic energy production and last the duration of the game (1)	4

Question Number	Answer	Mark
16	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. ATP-PC • It fuels activity lasting up to 10 seconds • Used for explosive power/high intensity activity • Phosphocreatine/PC/CP breaks down to release energy, which is transferred to ADP to resynthesize ATP	6

	• So Rhea will be able to sprint for a tackle and down the pitch • So Rhea will be able to generate power in her hits • Recovery time is 1:10/10sec-4 mins, so Rhea will run out of energy if she continues to work at this intensity Lactate • It fuels activity lasting 10 seconds – 3 minutes, when the ATP-PC system can no longer do it • It fuels medium/high intensity activity • Glucose/glycogen/carbohydrates is broken down to release/yield ATP/energy • Used when doing long sprints (e.g.) a recovery run from a counterattack after sprinting up the pitch • Recovery time is 3mins-2 hours, so Rhea will have to transfer to the aerobic system if she continues to work at this intensity General information • Anaerobic means that it does not require oxygen to produce energy • ATP-PC and lactate systems • Anaerobic activity is high intensity for a short duration	
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Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. Limited analysis which contains generic assertions rather than interrelationships or linkages.
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Question Number	Answer	Mark
17	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Respiratory system adaptations • Increased vital capacity, which will enable an increased and more efficient supply of oxygen to his working muscles • Increased strength of the respiratory muscles • The diaphragm and intercostal muscles increase in strength, allowing for greater expansion of the chest cavity • Therefore it is easier for him to take deeper breaths • Increased oxygen and carbon dioxide diffusion rates, allows oxygen and carbon dioxide to diffuse more rapidly • Thus supplying more oxygen to the working muscles and removing carbon dioxide quicker Muscular system adaptations • Muscular hypertrophy increasing his ability to generate more force/power • Increase in tendon strength, increasing his ability to generate more force/power • Increase in the size/number of mitochondria, which results in the muscles being able to produce more aerobic energy • Increase in myoglobin stores, therefore, more oxygenated blood can be transferred to the mitochondria and energy production will be increased • Increased storage of glycogen • Increased storage of fat • To use as an energy source to fuel him during the swim, when carbohydrates are low/intensity is low • Increased tolerance to lactate • Due to an extended capillary network, which enables greater volumes of blood to enter the muscle, more oxygen is available to break down lactic acid,	8

	Impact on performance • (e.g) Helps propel the body forward in the water • (e.g) Performance will be improved and fatigue will be delayed • (e.g) Therefore he will be able to swim at a higher intensity for longer duration • (e.g) Maintain technique • (e.g) Change pace in the race/swim a faster time • Accept any relevant swimming applied points,	
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1-3	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	4-6	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	7-8	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.



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Candidate surname					Other names					
Pearson BTEC Level 3 Nationals Certificate	Centre Number					Learner Registration Number				
Friday 15 January 2021										
Morning (Time: 1 hour 30 minutes)					Paper Reference 31524H					
Sport										
Unit 1: Anatomy and Physiology										
You do not need any other materials.								Total Marks		

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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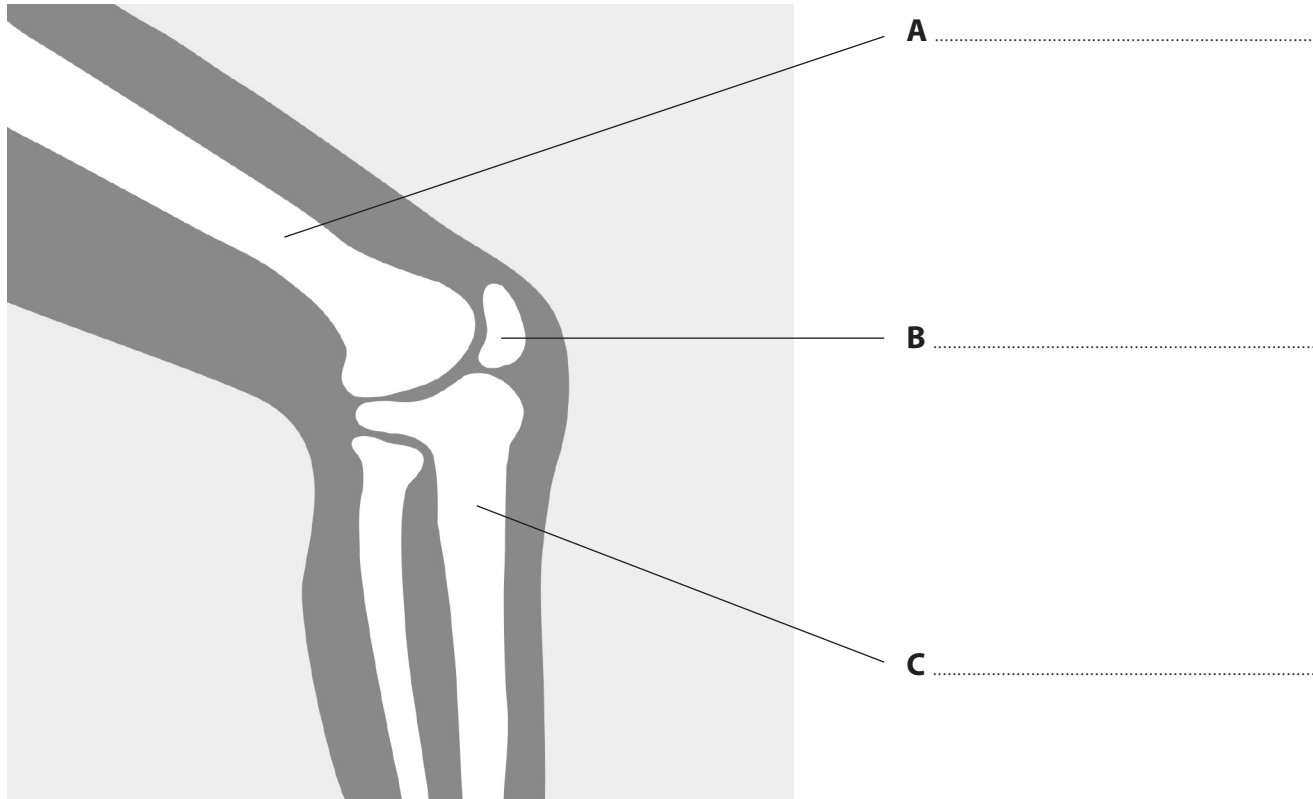
SECTION A: The Skeletal System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 1 shows the bones at the knee.

- 1 (a) Identify the bones labelled A–C in Figure 1.

(3)



Source: © Oleksandr Malysh/Shutterstock

Figure 1

Joints are classified according to the degree of movement they allow.
One classification is a cartilaginous (slightly moveable) joint.

- (b) Give **one** example of a cartilaginous (slightly moveable) joint.

(1)

(Total for Question 1 = 4 marks)



Rose is a road runner. Two of the functions of the skeleton are to provide leverage and weight bearing.

- 2 (a) Explain how the skeletal functions of leverage and weight bearing are used when Rose runs.

(4)

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The skeletal system has a number of functions other than leverage and weight bearing.

- (b) State **two other** functions of the skeletal system.

(2)

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(Total for Question 2 = 6 marks)



P 6 7 4 7 5 A 0 3 2 0

3 Explain the function of ligaments.

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(Total for Question 3 = 3 marks)

TOTAL FOR SECTION A = 13 MARKS

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SECTION B: The Muscular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 2 shows the muscles in the lower leg.

4 Identify the muscles labelled A–C in Figure 2.

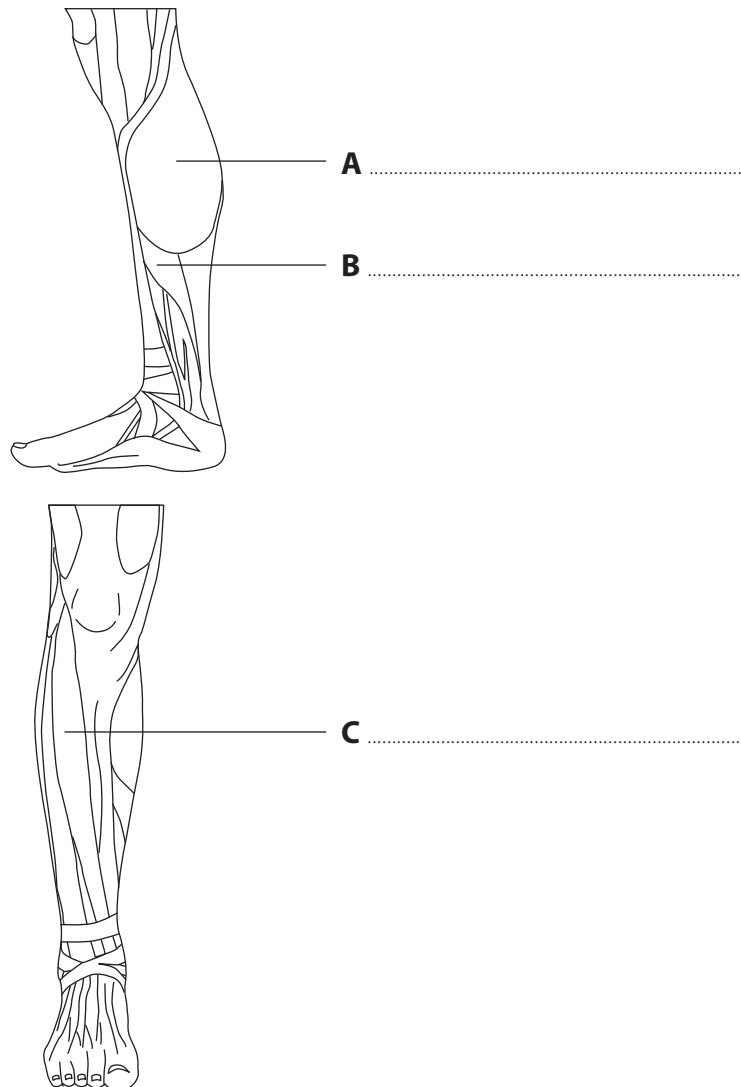


Figure 2

(Total for Question 4 = 3 marks)



5 State **three** characteristics of type I muscle fibres.

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(Total for Question 5 = 3 marks)

6 Describe the role of a synergist muscle.

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(Total for Question 6 = 2 marks)

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Marcellous is a sprinter.

Marcellous lifts heavy weights as part of his training regime.

- 7 (a) Explain the response of the muscular system from one session of lifting heavy weights.

(2)

Over a period of time Marcellous's muscular system has adapted to lifting the heavy weights. This lifting has caused muscular hypertrophy.

- (b) Explain the impact of muscular hypertrophy on Marcellous's sprinting performance.

(3)

(Total for Question 7 = 5 marks)

TOTAL FOR SECTION B = 13 MARKS



P 6 7 4 7 5 A 0 7 2 0

SECTION C: The Respiratory System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

- 8** Describe how the mechanism of breathing for **inspiration** changes during **exercise**.

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(Total for Question 8 = 4 marks)

- 9** Describe the role of the **internal** intercostal muscles during **expiration** when **exercising**.

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(Total for Question 9 = 2 marks)

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Dave is a tennis player. During the match he needs to supply his lungs with oxygen.

- 10** (a) Explain why gaseous exchange of oxygen is **faster** at the alveoli during exercise than at rest.

(4)

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(Total for Question 10 = 10 marks)

TOTAL FOR SECTION C = 16 MARKS

SECTION D: The Cardiovascular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 3 shows the heart.

11 Identify the structures of the heart labelled **A–D** in **Figure 3**.

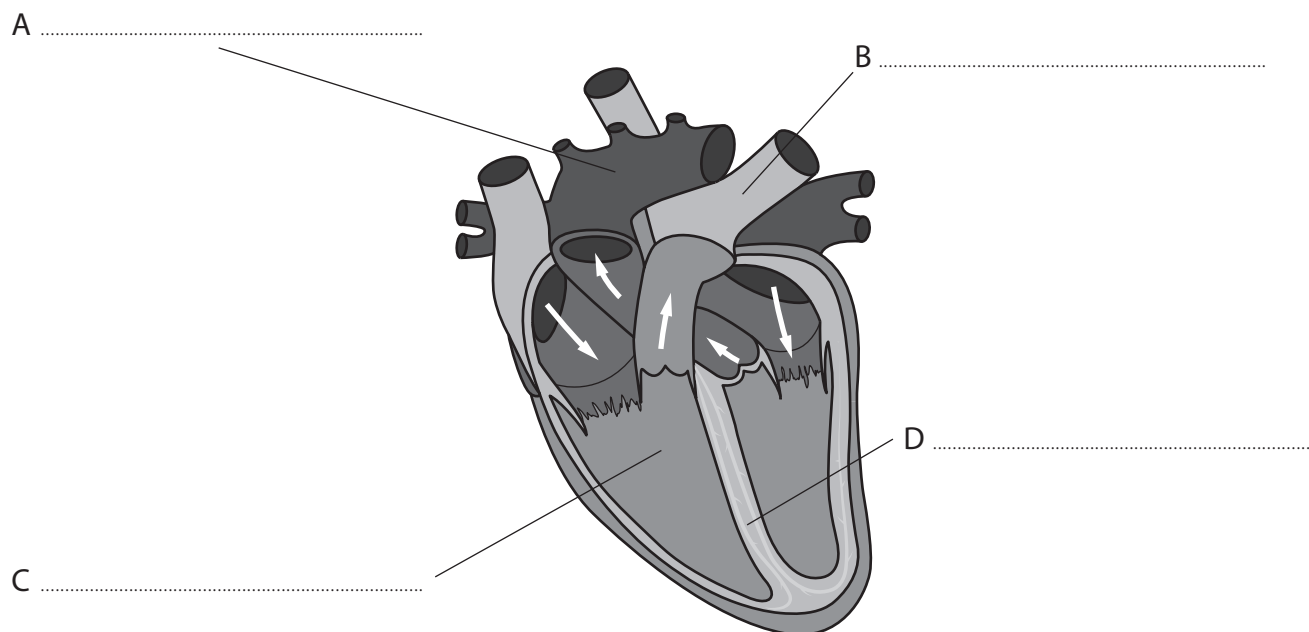


Figure 3

(Total for Question 11 = 4 marks)

12 Describe the role of semi-lunar valves.

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(Total for Question 12 = 2 marks)



P 6 7 4 7 5 A 0 1 1 2 0

Steph is a hockey player. She has been playing for several years and there have been cardiovascular adaptations in her body. One of these adaptations is an increase in blood volume.

13 (a) Explain how an increase in blood volume impacts on Steph's hockey performance. (4)

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(b) State what happens to Steph's stroke volume when she is playing hockey. (1)

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Steph's cardiovascular system has adapted as a result of playing hockey.

- (c) Analyse how cardiac hypertrophy and capillarisation of skeletal muscle and alveoli affect Steph's performance in a hockey match.

(6)

(Total for Question 13 = 11 marks)

TOTAL FOR SECTION D = 17 MARKS



P 6 7 4 7 5 A 0 1 3 2 0

SECTION E: Energy Systems for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 4 shows a graph of energy system contributions to exercise.

14 Identify the energy systems labelled A–C in Figure 4.

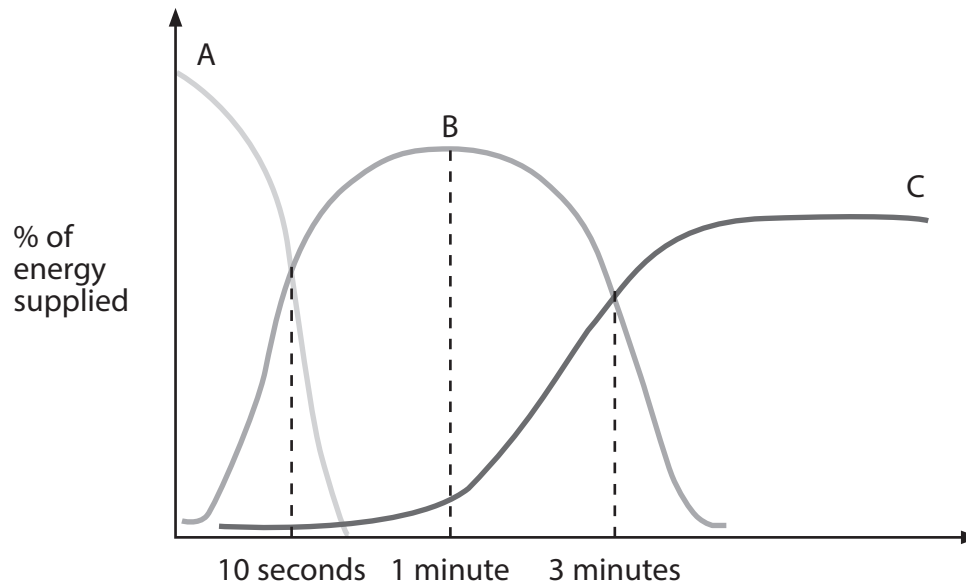


Figure 4

A

B

C

(Total for Question 14 = 3 marks)



Crystal is a long-distance cyclist. Crystal's aerobic system has adapted during her training regime. One of these adaptations is that Crystal can now use fats as an energy source more quickly.

15 Explain **two other** adaptations there have been in Crystal's aerobic energy system **and** the impact of these adaptations on her performance.

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(Total for Question 15 = 4 marks)



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(Total for Question 17 = 8 marks)

TOTAL FOR SECTION F = 8 MARKS
TOTAL FOR PAPER = 80 MARKS





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Mark Scheme (Results)

January 2021

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Unit 1: Anatomy and Physiology

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Question Number	Answer	Mark
1a	Award one mark for labelling each bone correctly. • A – Femur (1) • B – Patella (1) • C – Tibia (DNA Tibular) (1) Accept phonetic spelling. *DNA = Do not accept	3
1b	Award one mark for an example of a cartilaginous joint. • Vertebra(e)/cervical/thoracic/lumbar – (DNA sacrum/coccyx) • Ribs/sternum • Pubic symphysis Accept phonetic spelling. *DNA = Do not accept	1
2a	Award one mark for identifying why leverage and one mark for identifying why weight-bearing functions will help complete the training and one mark for explaining each reason. Leverage enables the bones to move (1) by providing sites for muscles to attach (1) The muscles pull on the bones (1) so the bones act as levers to move her legs (1) Weight bearing absorbs the shock (1) reducing the risk of injuries/shin splints (1) The skeleton is strong/can bear the weight of the <u>body</u> (1) and can hold the body upright (1) Accept any other appropriate answer.	4

2b	Award one mark for stating each function of the skeleton for a total of two marks. • Support/Supporting framework/shape (1) • Protection (1) • Attachment (for skeletal muscle) (1) • Blood Production/(Red) blood cell production/RBC production (1) • Reduce friction (across a joint) (1) Storage of minerals (1)	2
3	Award up to three marks for explaining the function of ligaments. Ligaments connect bone to bone (1) they provide strength/stability/support to the joint (1) by stopping any unwanted movement/reducing the risk of injury (1) Accept any other appropriate answer.	3
4	Award one mark for labelling each muscle correctly. • A – Gastrocnemius (1) (DNA Calf) • B – Soleus (1) • C – Tibialis anterior/Tibialis ant (1) Accept phonetic spelling. *DNA = Do not accept	3
5	Award one mark for each characteristic of type I muscle fibres, up to three marks. • Contract with little force (1) • Contract slowly (1) • Resistant to fatigue (1) • Small in size (1) • Red in colour (1) • Capillary density is high (1) • The number of mitochondria is high (1) • Oxidative (1) • High Myoglobin stores (1) Accept any other appropriate answer.	3

6	Award up to two marks for describing the role of a synergist. Supports (1) the agonist to carry out the movement (1) and stabilises the joint/ (1) Accept any other appropriate answer.	2
7 (a)	Award one mark for identifying the response to lifting heavy weights and one mark for linked expansion. Causes microtears in the muscle fibres (1) due to the muscle being overloaded (1) Accept any other appropriate answer.	2
7 (b)	Award one mark for explaining muscular hypertrophy and up to two further marks for linked expansion. Increase in size/strength of Marcellous's muscles (1) will enable him to generate more power/force (1) and run a faster time (1) Accept any other appropriate answer.	3
8	Award four marks for describing the mechanism of inspiration at exercise. - Diaphragm contracts/<u>external</u> intercostal muscles contract with <u>more force</u> (1) - Additional muscles are used/scalene/sternocleidomastoid/SCM/pectoralis minor (1) - The volume of the thoracic cavity increases <u>more</u>/ribs move <u>up and out more</u> (1) - The pressure in the thoracic cavity is reduced (1) <u>More</u> air comes in (1)	4
9	Award two marks for describing the role of the internal intercostal muscles when expiring when exercising. Contract (1) to bring the ribs down and in <u>quicker</u>/reduce the volume of the thoracic cavity <u>quicker</u> (1)	2

10a	Award up to four marks for explaining the process of gaseous exchange of oxygen at the alveoli during exercise. Muscles are using <u>more</u> O₂ (1) therefore a lower partial pressure/ppO₂ in the capillaries near the alveoli (1). There is an <u>increased</u> breathing rate (1), which means more O₂ /higher partial pressure/ppO₂ is in the alveoli (1) which creates a steeper diffusion gradient (1) and O₂ moves quicker (from the alveoli) into the blood/capillaries (1)	4
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10(b)	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • Increased strength of respiratory muscles/Diaphragm and intercostal muscles • Will increase the force of contraction of respiratory muscles • Allowing for a greater expansion of the chest cavity/the lungs to expand more/increasing tidal volume • Increased Vital capacity • Maximum amount of air exhaled from the lungs following deepest breath/IRV + ERV +TV • Which enables more oxygen to be inhaled • Therefore greater volume of air expired • Increase in Oxygen & Carbon Dioxide diffusion rates • Increases the amount of oxygen that is diffused into the blood • Increases the amount of carbon dioxide that is diffused out of the body Impact on Performance • Dave can work at a higher intensity for longer/fatigue delayed, (e.g.) can maintain his shots and movement around the court late into the game • the delay of OBLA/lactate threshold is a result of Dave’s increased VO2max, Accept any other appropriate answer.	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question.

		Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	- Demonstrates some accurate knowledge and understanding. - Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. - Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	- Demonstrates mostly accurate knowledge and understanding. - Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. - Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

11	Award one mark for labelling each structure of the heart correctly. - A – Aorta (1) - B – Pulmonary artery/PA (1) - C – Right ventricle/RV (1) - D – Septum (1) Accept phonetic spelling.	4
12	Award up to two marks for describing the role of the semi-lunar valves. Control the flow of blood from the ventricles/to the aorta/to the pulmonary artery (1) prevent backflow (1)	2
13 (a)	Award two marks for explaining the impacts of increased blood volume and two marks for linked justification. (More blood increases) the ability to transfer <u>more</u> oxygen to the working muscle (1) and remove carbon dioxide (1) therefore enabling Steph to play at a higher intensity for a longer duration (1) and delay the onset of fatigue (1) Accept any other appropriate answer	4

13 (b)	Award one mark for stating the change in stroke volume in the game. Increases/Goes up/Higher/Greater (1)	1
13 (c)	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Cardiac Hypertrophy • Cardiac hypertrophy is when the heart gets bigger and the left ventricle wall gets thicker • This results in a larger/more forceful contraction • Therefore SV will increase • Because the heart will be able to hold more blood • And cardiac output will increase during exercise • At rest, heart rate will decrease meaning that the heart will not need to work as hard Capillarisation • Increase in the number and size of blood vessels • More blood will go to the working muscles • More efficient delivery of oxygen and nutrients • Removal of waste products/CO₂ Impact on performance • Fatigue will occur later in the game • Maintaining a faster pace/higher intensity throughout the match (e.g. can run down the wing and recover)	6

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

14	Award one mark for labelling each part of the graph correctly. • A – ATP-PC (1) • B – Lactate/lactic acid/LA (1) • C – Aerobic (1)	3
15	Award one mark for each aerobic adaptation and one mark for linked expansion/justification.	4

	Increased storage of glycogen (1) enabling Crystal to keep cycling at a high intensity/complete races in a quicker time (1) Increased number of mitochondria (1) increases the ability to produce more (aerobic) energy and last the duration of the race (1)	
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16	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. - ATP-PC is the <u>main</u> system used for the long jump - The long jump is an anaerobic activity - It fuels activity lasting up to 8–10 seconds - Uses explosive power/high intensity activity - Used when sprinting in the run up/jump phase - PC is broken down to release energy - Phosphocreatine stores are replenished in 2–4 minutes. Recovery time is short - Boris will be recovered ready for the next round	6
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Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 1	1–2	- Demonstrates isolated elements of knowledge and understanding. - Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. - Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	- Demonstrates some accurate knowledge and understanding. - Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. - Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.

Level 3	5-6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.
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17	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Muscular • Increased blood supply to the muscle • Increased pliability/elasticity of muscle tissue • Increases temperature of muscle • Decreases muscle viscosity • Increases speed and strength of contraction • This all results in increased flexibility and therefore reduces risk of injury muscle strains/pulls/tears in the game Cardiovascular • Increases heart rate/increases stroke volume/increases cardiac output/$Q=SV \times HR$ • Increases blood pressure to pump blood faster • Increases blood temperature, which reduces blood viscosity, which allows blood to get around the body quicker. • Therefore, <u>more</u> oxygenated blood to muscles and increased removal of carbon dioxide and waste products out of the blood stream • Activates vascular shunt mechanism/redirection of blood • Which redistributes blood from non-essential organs to the working muscles	8
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	• Therefore, <u>more</u> oxygen can be supplied to sustain energy production during the netball game • Increases diffusion of oxygen from haemoglobin to muscles, which causes a steeper diffusion gradient and increased diffusion rate of oxygen into the blood stream • Enables the players to not experience the early feeling of being out of breath, because the heart rate is elevated to similar to the rate in the game	
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Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	4–6	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	7–8	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.



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Please check the examination details below before entering your candidate information

Candidate surname					Other names					
Pearson BTEC Level 3 Nationals Certificate	Centre Number					Learner Registration Number				
Tuesday 21 May 2019										
Morning (Time: 1 hour 30 minutes)					Paper Reference 31524H					
Sport										
Unit 1: Anatomy and Physiology										
You do not need any other materials.								Total Marks		

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION A: The Skeletal System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

Figure 1 shows the bones of the upper body.

1 Identify the bones labelled **A-C**.

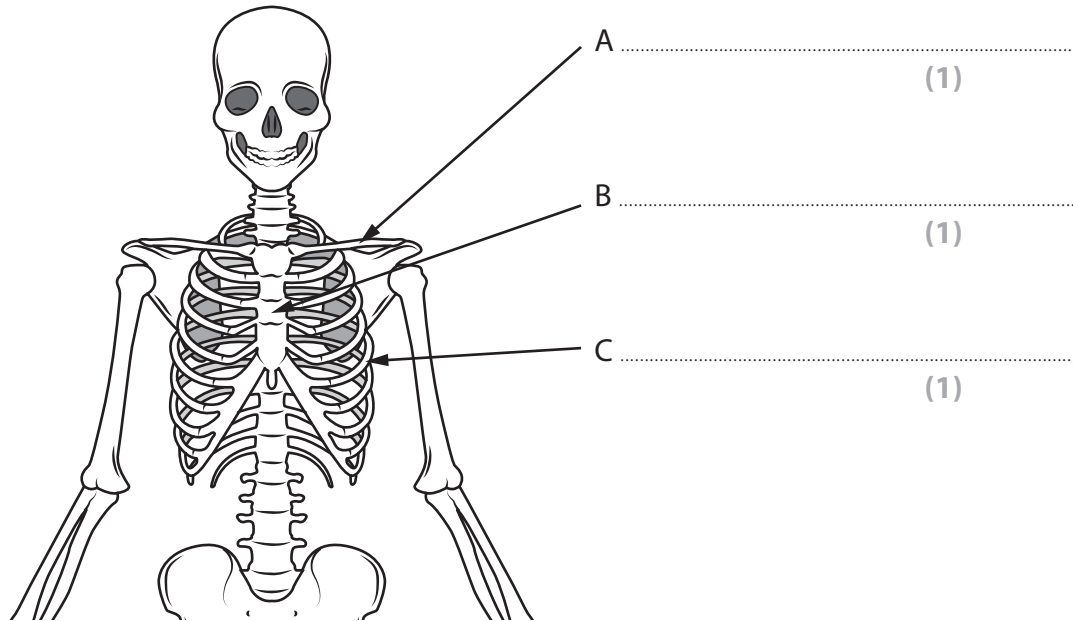


Figure 1

(Total for Question 1 = 3 marks)

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Joints are classified by how much movement they allow.

One category of classification is a fibrous joint.

2 (a) Give an example of a fibrous joint.

(1)

(b) State **two** characteristics of a fibrous joint.

(2)

1

2

(Total for Question 2 = 3 marks)



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Alice wanted to start resistance (weight) training, but has been told by her doctor that she is too young to do so.

(b) Explain why resistance (weight) training is not recommended for children.

(3)

(Total for Question 3 = 7 marks)

TOTAL FOR SECTION A = 13 MARKS



P 6 2 8 7 7 A 0 5 2 4

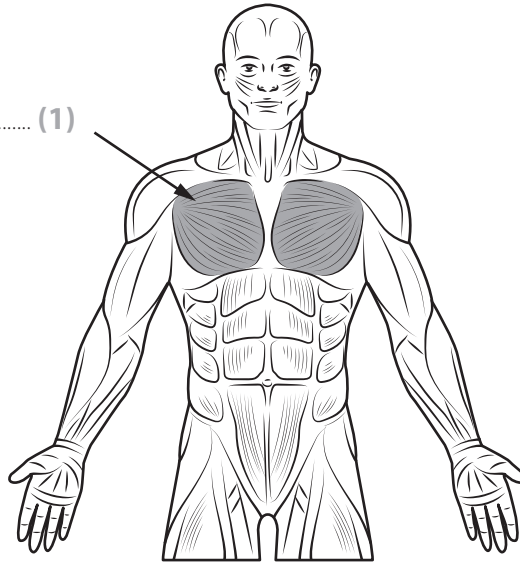
SECTION B: The Muscular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

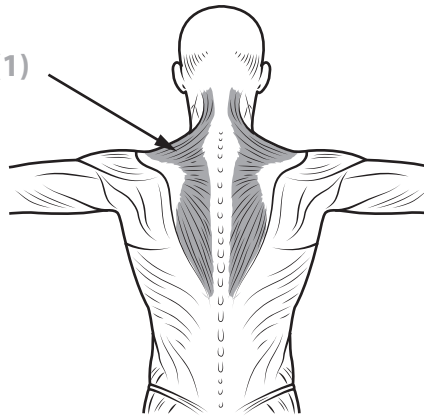
Figure 2 shows the muscles in the upper body.

4 Identify the muscles labelled A-C.

A (1)



B (1)



C (1)

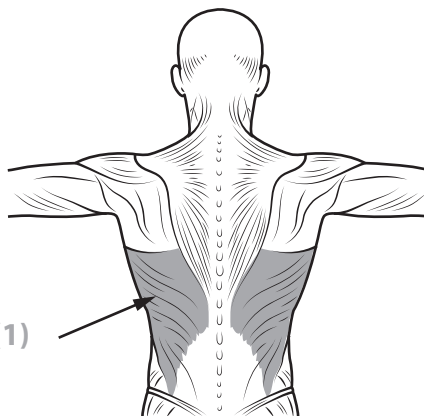


Figure 2

(Total for Question 4 = 3 marks)

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5 State **three** characteristics of type IIx muscle fibres.

1

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2

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3

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(Total for Question 5 = 3 marks)

6 State **two** characteristics of smooth muscle.

1

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2

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(Total for Question 6 = 2 marks)



P 6 2 8 7 7 A 0 7 2 4

Julie is a marathon runner.

Julie has trained for many years so her body has adapted to increase the storage of muscle glycogen.

- 7 (a) Explain why an increased storage of muscle glycogen benefits Julie during a race.

(2)

Julie is on a training run. One response to the run is that the temperature of her muscles increases.

- (b) Explain what causes muscle temperature to increase during exercise.

(3)

(Total for Question 7 = 5 marks)

TOTAL FOR SECTION B = 13 MARKS

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Answer ALL questions. Write your answers in the spaces provided.

[illegible]

9 (a) State the meaning of the term 'vital capacity'.

(b) State the effect that altitude has on a person's breathing rate.



Shelly is a cross country runner. During a training run her tidal volume increases.

10 (a) Explain the effects an increased tidal volume would have on Shelly’s cross country performance.

(4)

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During the training session, Shelly's breathing rate increases.

(b) Analyse how Shelly's increased breathing rate is controlled.

(6)

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(Total for Question 10 = 10 marks)

TOTAL FOR SECTION C = 16 MARKS



SECTION D: The Cardiovascular System for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

11 Name the structures **A–D** described in **Table 1**.

Structure	Description
A-	The body’s main artery, originating from the left ventricle and carries oxygenated blood.
B-	A blood vessel that receives de-oxygenated blood from the body to empty into the right atrium.
C-	Blood vessel that supplies oxygenated blood to the heart muscle.
D-	Prevents backflow of blood into the ventricle.

Table 1

(Total for Question 11 = 4 marks)

12 Describe sudden arrhythmic death syndrome (SADS).

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(Total for Question 12 = 2 marks)



Imran is a rugby player. He has been playing for several years and his body has undertaken cardiovascular adaptations. One of these adaptations is a decreased heart rate recovery time.

- 13 (a) Explain how a decreased heart rate recovery time benefits Imran's rugby performance.

(2)

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Imran is playing in a rugby match.

In the match, Imran's cardiac output changes from rest.

- (b) (i) State what happens to Imran's cardiac output during a rugby game.

(1)

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- (ii) State the **two** cardiovascular responses of the body that cause this change in cardiac output.

(2)

1

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Figure 3 shows Imran's distribution of blood flow at rest and during the rugby game.

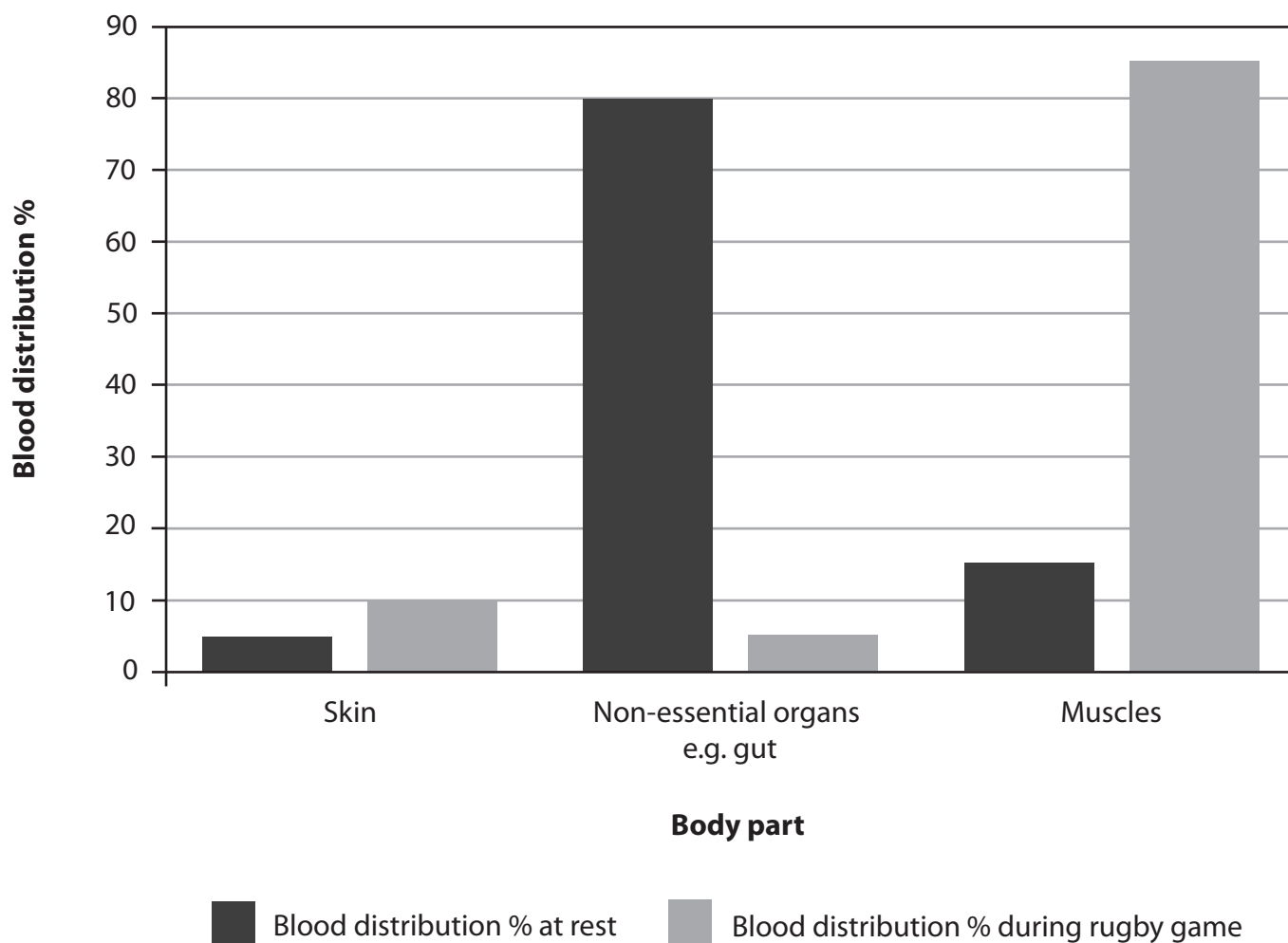


Figure 3

(c) Analyse, using **Figure 3**, the changes in distribution of Imran's blood flow to his skin, non-essential organs and muscles when playing rugby.

(6)

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(Total for Question 13 = 11 marks)

TOTAL FOR SECTION D = 17 MARKS



SECTION E: Energy Systems for Sports Performance

Answer ALL questions. Write your answers in the spaces provided.

14 Describe the process of the electron transport chain.

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(Total for Question 14 = 4 marks)



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(Total for Question 16 = 6 marks)

TOTAL FOR SECTION E = 13 MARKS



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(Total for Question 17 = 8 marks)

TOTAL FOR SECTION F = 8 MARKS
TOTAL FOR PAPER = 80 MARKS



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Mark Scheme

June 2019

Pearson BTEC Level 3 - Sport

Unit 1: Anatomy and Physiology
(31524)

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June 2019

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Unit 1: Anatomy and Physiology – sample marking grid

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Marking grids should be applied positively. Learners must be rewarded for what they have shown they can do, rather than be penalised for omissions.
- Examiners should mark according to the marking grid, not according to their perception of where the grade boundaries may lie.
- All marks on the marking grid should be used appropriately.
- All the marks on the marking grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the marking grid.
- Where judgement is required, a marking grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the marking grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Mark
1	Award one mark for labelling each bone correctly. • A – Clavicle (1) (DNA collarbone) • B – Sternum (1) (DNA breastbone) • C – Ribs/Rib Cage (1) Accept phonetic spelling. • *DNA = Do not accept	3

Question Number	Answer	Mark
2a	Award one mark for an example of a fibrous joint. • Cranium – (DNA skull) • Pelvis – (DNA hips) • Sacrum – (DNA spine/vertebrae) • Coccyx – (DNA spine/vertebrae) Accept phonetic spelling. *DNA = Do not accept	1

Question Number	Answer	Mark
2b	Award one mark for each characteristic of a fibrous joint, up to a maximum of two marks. • Do not move/no movement (1) • Bones interlock/overlap (1) • The joints are held together with bands of fibrous/tough tissue (1)	2

Question Number	Answer	Mark
3 (a)	Award one mark for identifying why regular weight-bearing activities will help reduce osteoporosis and up to three marks for justifying that reason. The weight-bearing activities causes additional stress or damage to the bone/help make the bones stronger/denser (1) enabling osteoclasts to clear away old bone (1) increasing osteoblast activity to build new bone (1) allowing the bones to take up more calcium/minerals (1) and making them less likely to break/fracture in later life (1) Accept any other appropriate answer.	4

Question Number	Answer	Mark
3 (b)	Award up to three marks for explaining why weight training is not recommended for children. A child's bones are still growing (1) and too much force can damage the epiphyseal/growth plates (1) this can result in stunted growth/can stop growth (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
4	Award one mark for labelling each muscle correctly. • A – Pectorals/Pectoralis/Pectoralis Major/Pectoralis Minor (1) (DNA Pecs) • B – Trapezius (1) (DNA Traps) • C – Latissimus dorsi (1) (DNA Lats) Accept phonetic spelling. *DNA = Do not accept	3

Question Number	Answer	Mark
5	Award one mark for each characteristic of type IIX muscle fibres, up to three marks. • Contract with a lot of force (1) • Contract quickly (1) • Not resistant to fatigue/tire quickly (1) • Large in size (1) • White colour (1) • Capillary density is low (1) • The number of mitochondria is small/few (1) • Glycolytic capacity is high (1) • Anaerobic (1) • Phosphocreatine levels are high (1) Accept any other appropriate answer.	3

Question Number	Answer	Mark
6	Award one mark for each characteristic of smooth muscle, up to maximum of two marks. • Involuntary (contractions) (1) • Slow speed of contraction (1) Accept any other appropriate example.	2

Question Number	Answer	Mark
7 (a)	Award one mark for identifying why an increase in glycogen is beneficial to performance and one mark for linked expansion. Provide readily available energy in the muscle (1) enabling her to run faster for longer/work at a higher intensity for longer/withstand fatigue for longer (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
7 (b)	Award one mark for explaining what causes muscle temperature to increase and up to two further marks for linked expansion. Temperature increases because muscles need energy (1) fats and carbohydrates are broken down/breakdown of fuel/food (1) and produce heat as a waste product (1) Do not award – any reference to flexibility, Accept any other appropriate answer.	3

Question Number	Answer	Mark
8	Award four marks for describing the mechanism of expiration at rest. • Diaphragm relaxes / <u>external</u> intercostal muscles relax (1) • The volume of the thoracic cavity decreases/ribs move <u>down and in</u> (1) • The pressure in the thoracic cavity is increased (1) • and air comes out (1)	4

Question Number	Answer	Mark
9 (a)	Award one mark for providing a definition of vital capacity. • <u>Maximum</u> amount of <u>air</u> expired after <u>maximum</u> inhalation (1) Do not accept - oxygen/carbon dioxide.	1

Question Number	Answer	Mark
9(b)	Award one mark for stating the effect. • It increases/speeds it up/makes it faster	1

Question Number	Answer	Mark
10 (a)	Award two marks for identifying the effects and two marks for linked expansion. More CO₂ leaves the body (1) and more oxygen taken in (to the working muscles) (1). These maintain the intensity/speed of her running (1) and delay fatigue/enabling her to run a faster time (1) Accept any other appropriate answer.	4

Question number	Indicative content	
10(b)	Answers will be credited according to the learner’s demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • Can be controlled through neural and chemical control • Chemoreceptors detect a change in the partial pressure of the gases in Shelly’s body • Low levels/low partial pressure of oxygen/O2 • High levels/high partial pressure of carbon dioxide/CO2/low pH • These messages are sent to the medulla oblongata as the respiratory centre/RCC in the brain • Messages are then relayed to the respiratory muscles • Causing the diaphragm and intercostal muscles to work faster/harder • Enabling more oxygen to enter the lungs/bloodstream • More carbon dioxide to be removed from the bloodstream/body • That enables Shelly’s muscles to receive more oxygen and work aerobically. Accept any other appropriate answer.	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	5–6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

Question Number	Answer	Mark
11	Award one mark for naming each structure correctly. • A – Aorta (1) • B – Vena cava (1) • C – Coronary artery (1) • D – Semi-lunar/aortic/pulmonary valve(s) (1)	4

Question Number	Answer	Mark
12	Award up to two marks for describing sudden arrhythmic death syndrome. Disruption to the rhythm of the heart (1) which can cause the heart to stop beating / which could lead to death (1)	2

Question Number	Answer	Mark
13 (a)	Award one mark for the effects decreased heart rate recovery time has on performance, up to two marks. This allows Imran to recover following periods of (high intensity) work quickly (1) and therefore able to tackle/run again quicker (1) Accept any other appropriate answer.	2

Question Number	Answer	Mark
13 (bi)	Award one mark for stating the change in cardiac output correctly. • It increases (1)	1

Question Number	Answer	Mark
13 (bii)	Award one mark for identifying each component correctly, for a maximum two marks • Anticipatory rise (1) • Heart rate/HR (1) • Stroke volume/SV (1)	2

Question Number	Answer	Mark
13 (c)	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • Vasoconstriction of blood vessels/arterioles to areas where blood is not required • Vasodilation of blood vessels/arterioles to areas where blood is required, which results in increased blood flow to skeletal muscle • More blood is required to supply oxygen and nutrients to working muscles • Increased blood supply to remove waste products • Increased blood flow to the skin for thermoregulation to cool the body via evaporation of sweat • Less blood flow to the non-essential organs as the focus during exercise is for increased blood flow to support the body with movement rather than digestion • Redistribution of blood is called the vascular shunt mechanism • Redistribution of blood is controlled by the medulla	6

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates isolated elements of knowledge and understanding. • Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	3–4	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.

Level 3	5-6	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.
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Question Number	Answer	Mark
14	Award one mark for each descriptive point, up to a maximum of four marks. • (The electron transport chain/ETC) only occurs when oxygen is present (1) • It occurs in the mitochondria (1) • It receives hydrogen ions/hydrogen from the Krebs cycle (1) • 32-34 ATP are created (1) • Water is the by-product (1) Accept an ATP value between 32 to 34. Accept any other appropriate answer.	4

Question Number	Answer	Mark
15	Award one mark for each descriptive point, up to a maximum of three marks. • Glycogen/glucose is converted into pyruvate/pyruvic acid (1) • There is oxygen present (1) • The pyruvate/pyruvic acid goes into the Krebs cycle (1) • This reaction yields 2 ATP molecules (1) • This takes place in the sarcoplasm (1)	3

Question Number	Answer	Mark
16	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. • All three systems work together to provide the energy. ATP-PC system • System used for activity lasting 8-10 seconds and the main system used in tennis • Used when hitting powerful shots/smash/serve • Sprinting around the court to return shots • Phosphocreatine stores are replenished in 2-4 minutes, so supported during breaks between points and games. Lactate system • Provides energy for moderate to high intensity activity • Only lasts for 1-3 minutes • Will be used during long, intense rallies that last longer than 10 seconds • When recovery periods between points are short. Aerobic system • Submaximal system in exercise longer than 1 minute • Helps support the anaerobic systems during the game • To supply the energy when the match is long • Works when intensity falls between points/games and helps the muscles recover.	6
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor

Level 1	1-2	• Demonstrates isolated elements of knowledge and understanding. • Provides little or no reference to the context in the question. • A conclusion may be presented, but will be generic and the supporting evidence will be limited. Limited attempt to address the question. • Response is likely to lack clarity, organisation and the required technical language.
Level 2	3-4	• Demonstrates accurate knowledge and understanding. • Line(s) of argument occasionally supported through the application of relevant references to context in question. • Judgement is made from a partially-developed discussion, although the discussion may be imbalanced or superficial in places. Learners will produce some statements with development in the form of mostly accurate and relevant factual material leading to an assessment being presented. • The response may contain parts which lack clarity or organisation. There is evidence of correct technical language being used.
Level 3	5-6	• Demonstrates accurate knowledge and understanding. • Line(s) of argument supported throughout by sustained application of relevant references to context in the question. Might demonstrate the ability to integrate and synthesise relevant systems. • Arrives at a supported judgement from a well-developed and logical balanced discussion, containing logical chains of reasoning. Demonstrates an awareness of competing arguments using these to reach a valid assessment. • Response demonstrates good organisation, clarity and use of technical language.

Question Number	Answer	Mark
17	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and level descriptors below. The indicative content that follows is not prescriptive. Answers may cover some/all of the indicative content but should be rewarded for other relevant answers. Cardiovascular system • Callum's cardiovascular system will have adapted through cardiac hypertrophy • this increases stroke volume and reduces the need for the heart to beat so many times thus slowing the cardiac cycle • this will give Callum a greater heart rate range to utilise when playing hockey • giving him a greater maximal cardiac output • Callum will also have an increased blood volume • meaning more oxygen is transported to his muscles • More efficient and his recovery time will be reduced • Capillarisation around the alveoli and in the muscles will increase the efficiency of gaseous exchange Respiratory system • Callum's respiratory muscles will increase in strength • contributing to an increased vital capacity to increase the volume of air moving into and out of the lungs • this means that he will be able to better utilise the increased oxygen coming into the body (through increased lung volumes) to sustain aerobic energy production for longer • so he is able to play hockey for sustained periods • there will be an increase in oxygen and carbon dioxide diffusion rates • Callum's increased ability to utilise oxygen breathed into the body will mean that more oxygen is available to break down lactic acid Application to Performance • e.g. Maintain man marking for longer • e.g. Recover from a run down the wing quicker • e.g. Make more repeated sprints to chase the ball • the delay of OBLA/lactate threshold is a result of Callum's increased VO₂max, i.e. the maximal amount of oxygen Callum is able to utilise per minute, so we can see that Callum's continuous training sessions resulted in an increased VO₂ max, increasing Callum's fitness for hockey	8

Mark scheme (award up to 8 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding.

		• Breaks the situation down into component parts and a few of the points made will be relevant to the context in the question. • Limited analysis which contains generic assertions rather than interrelationships or linkages.
Level 2	4-6	• Demonstrates some accurate knowledge and understanding. • Breaks the situation down into component parts and some of the points made will be relevant to the context in the question. • Displays a partially developed analysis which considers some interrelationships or linkages but not always sustained.
Level 3	7-8	• Demonstrates mostly accurate knowledge and understanding. • Breaks the situation down into component parts and most of the points made will be relevant to the context in the question. • Displays a developed and logical analysis which clearly considers interrelationships or linkages in a sustained manner.

